

6.1 DDQ - Design, Develop and Qualify the Solution (FR-CSF-EN - L.2.2.0)Reference : **87202127-00-DDQ-GRP-EN**Release : **L.2.2.0**

Printed on 30/11/2023 from Chorus 2.0 Portal

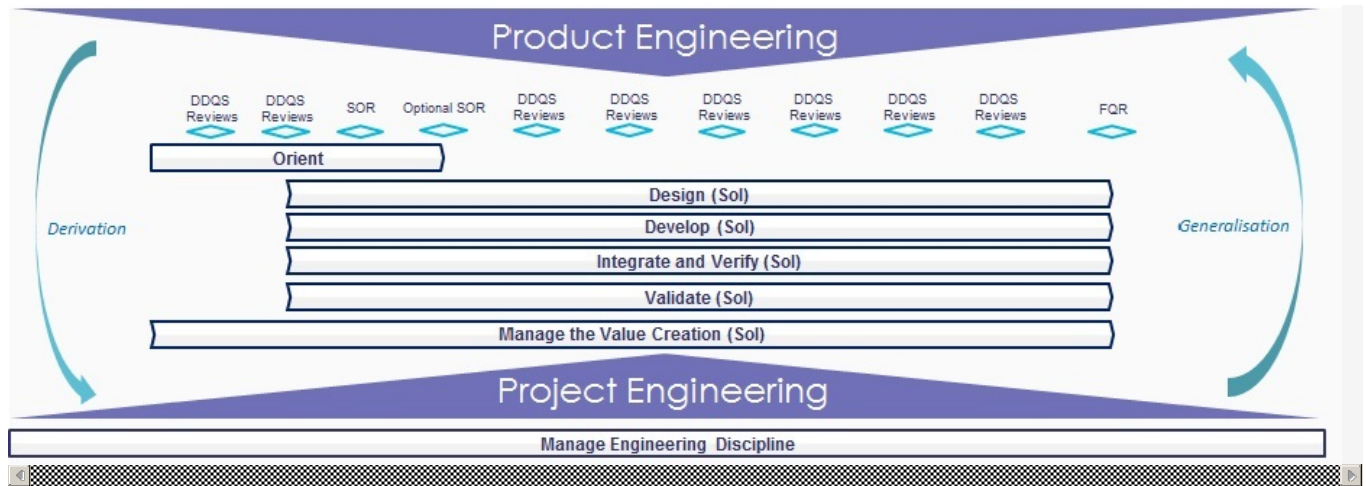
Profile : **France / THALES / EN****Table of contents****Design, Develop and Qualify the Solution**

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6.1 DDQ - Design, Develop and Qualify the Solution (ref: 87202127-00)

Activity flow diagram



Presentation

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Objective

To provide products and customer solutions in consistency with all stakeholders' needs and entity policies , as well as continuously enhance engineering competitiveness.

Description

This process encompasses activities on bid, products and projects (i.e. Orient, Design, Develop, Integrate & Verify, Validate, Manage the Value Creation) and a transverse one (i.e. Manage the Engineering Discipline).

Context

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This process is applicable to a solution or solution element like a system, sub-system, equipment or a component (building block or lower level component) , as well as to any associated enabling system. It is not applicable to partners or subcontractors, except otherwise agreed.

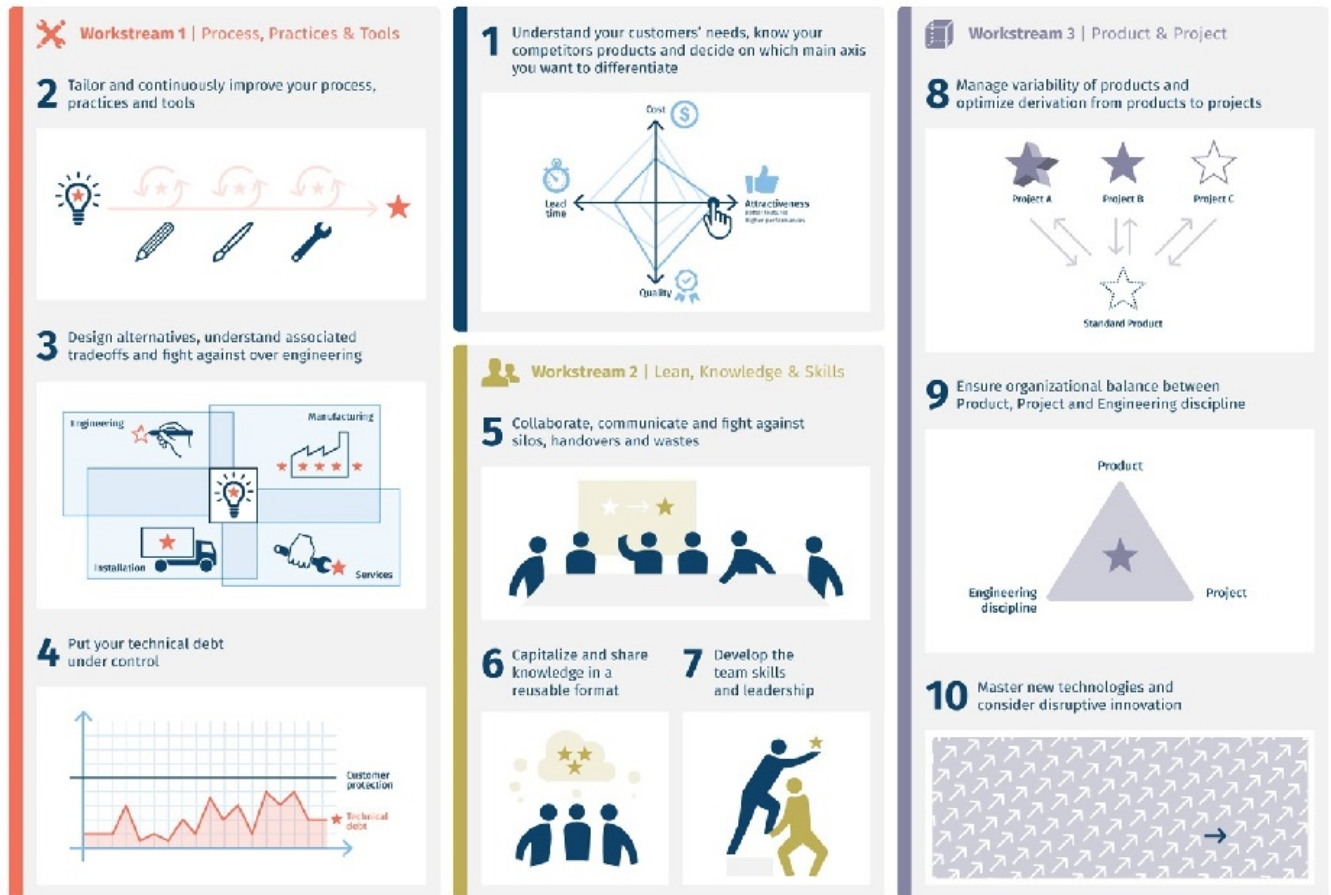
Rules

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0105: Engineering Golden Rules

Engineering activities shall be performed with the objective of permanently fulfilling the 10 Golden rules of Engineering Competitiveness.

★ The 10 Golden rules of Engineering Competitiveness



0110: DDQ - TCIS:

All purchased catalogue items , COTS & Open Source (OSS) being part of the design engineering solutions and products shall be registered in the Thales Component Information System (TCIS). This includes the followings :

Hardware electrical & electronic components, raw material & mechanical components, chemical products substances & process, optical components.

Hardware COTS (power supply, electronic boards....)

Software COTS & OSS (DBMS, UIMS, OS....).

0115: Engineering documents management (Solution)

In consistency with document management process the control rules for engineering documents are the following:

- Roles of writer, verifier and approver are defined for main engineering documents in tables below.
- Even when roles are cumulated the writer, the verifier and the approver shall be at least two different persons.

These rules may be superseded at entity level to better fit with business constraints (e.g. for certification purpose).

SOLUTION	Written by	Verified by	Approved by
D&JF (archit. choices)	ARC (SE ¹)	EDM	PDA (ARC ¹)
OCD	ARC	EDM	PDA (ARC ¹)
DVa	EDM & PDA	_4	DA
D&JF (conformity to requirements.)	SE ²	EDM	PDA (ARC ¹)
SSS , ICD	SE ²	EDM	PDA (ARC ¹)
STD/STR	SE ²	EDM	IVVQM
D&JF (Design choices)	ARC (SE ¹)	EDM	PDA (ARC ¹)
SSDD	SE ³	EDM	PDA (ARC ¹)
IVVQP	IVVQM	QAM	EDM & PDA(ARC ¹)
			HoD & PDA(ARC ¹) & PM ³
TITLE / REFERENCE	EDM TYPE	SCOPE	QAM SPECIFIC
			LANG UAGE
			DATE

Documents

Booklet 6.1 - Design Development and Qualify the Solution
87212268-DDQ-GRP-EN-002
<https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=213599806&vernum=8>

1: solution element case or if no PDA at solution level (e.g. for B0/A0 Bids/Projects, in accordance with Main Rules for Design Authority Operation 87201633-DDQ-GRP-EN-002)

2: several authors may be indicated

3: On upper level EDM in case of a solution element

4: Chief Engineer when appointed

CC Metrics KPI
87216428-DDQ-GRP-EN-002
<https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=213599806&vernum=8>

Instru
ction
EN 27/06/2023

In accordance with the new General Data Protection Regulation (GDPR) of the European commission, a solution who processes personal data must comply with the "Privacy by Default" and "Privacy by Design" principles by defining requirements e.g. in system engineering plan and specifications and then implementing and verifying them in the proposed solution and its elements.

<https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=191043949&vernum=8>

DIS Engineering KPIs Instruction
65734407-DDQ-DIF-EN-002
<https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=219988200&vernum=3>

Instru
ction
DIS
EN 25/07/2023

Discovering Thales Engineering Complex Distributed Systems
87215608-DDQ-GRP-EN-002
<https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=213567426>

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Discovering Thales Engineering Equipments
87215609-DDQ-GRP-EN-002
<https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=213505839>

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EN 27/06/2023

Discovering Thales Engineering Software Based Solutions
87215610-DDQ-GRP-EN-002
<https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=214937656>

Guid
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Softw
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EN 27/06/2023

EMS tailoring
87201604-DDQ-GRP-EN-015
<https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=96089427&vernum=18>

Instru
ction
EN 28/11/2023
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Engineering governance-Redirection
87210749-DDQ-GRP-EN-003
<https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=119577451&vernum=4>

Instru
ction
EN 23/03/2021

Engineering reviews instruction
87211053-DDQ-GRP-EN-006
<https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=119567360&vernum=9>

Instru
ction
EN 11/07/2023

Engineering function data policy
87218469-DOC-GRP-EN-001
<https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=268723956>

Guid
eline
EN 14/03/2023

IVVQ Handbook
87100124-DDQ-GRP-EN-009
<https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=19448365>

Guid
eline
EN 27/06/2023

TITLE / REFERENCE	TYPE	SCOPE	SPECIFIC	LANG UAGE	DATE
Instruction for producing Design & Justification File (D&JF) 87201508-DDQ-GRP-EN-009 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=96091671&vernum=12	Instru ction		Hard ware Softw are Soluti on	EN	28/11/2023 U!
Leaflet 6.1 DDQS presentation 87210246-DDQ-GRP-EN-011 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=133597280	TRS Traini ng			EN	27/06/2023
Main Rules for Design Authority Operation (MRDAO) - redirection 87201633-DDQ-GRP-EN-005 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=96080836&vernum=4	Instru ction			EN	15/01/2019
Milestones reviews awareness 87214871-DDQ-GRP-EN-004 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=191296430	TRS Traini ng			EN	27/06/2023
Product Line Engineering (PLE) 87210086-DDQ-GRP-EN-008 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=96019975&vernum=10	Instru ction			EN	28/11/2023 U!
Product Safety Management 65734411-DDQ-DIF-EN-001 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=231746580&vernum=1	Instru ction	DIS		EN	22/02/2022
Product_Line_Engineering_PLE_Handbook 87208865-DDQ-GRP-EN-003 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=109677387	TRS Guid eline		Hard ware Softw are Soluti on	EN	27/06/2023
RT04 - 6.1 - DDQS training kit 87205192-DDQ-GRP-EN-018 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=20228801	TRS Traini ng			EN	27/06/2023
Safety Assurance 87211146-DDQ-GRP-EN-004 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=271355589&vernum=1	Guid eline			EN	25/04/2023
Safety Engineering Instruction 87204905-DDQ-GRP-EN-007 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=96088738&vernum=9	Instru ction		Hard ware Softw are Soluti on	EN	27/06/2023
UX Design in Engineering 87216348-DDQ-GRP-EN-001 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=218307077&vernum=3	Instru ction		Soluti on	EN	28/09/2021
VID 6.1 - Agile Software development in Thales 87211993-DDQ-GRP-EN-001 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=126611220	TRS Traini ng			EN	29/06/2017
eLearning 6.1 - Understand - Design, Develop and Qualify a Solution (6.1) 87205074-DDQ-GRP-EN-002 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=98480077	TRS Traini ng			EN	23/11/2017
ecodesign in engineering 87219150-DDQ-GRP-EN-001 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=283564050&vernum=1	Instru ction			EN	28/11/2023 N!

Inputs/Outputs

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Inputs

Configuration Baselines
Current technological and production constraints
Customer requirements, requests and needs
Customers, suppliers and partners feedbacks and third party complaints
Internal Executive summary - BKO
List of concepts/technologies validated by a main gate
Product Line strategy with roadmap
Project baseline
Purchasing strategy
SBP
Service Definition Document

Outputs

Engineering plans
Product Structures
Product data Package (PDP)
Qualified Solution
Statement of Work (SOW)
Technical solution

Responsible

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► Responsible ► Contributor

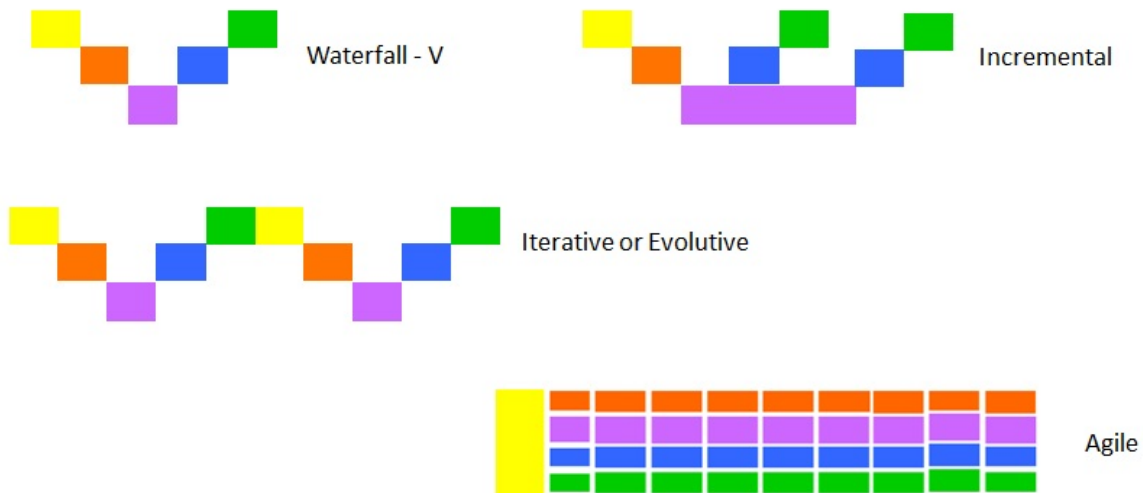
► Engineering Delivery Manager (EDM)	► Project Design Authority (PDA)
► Airworthiness Certification Engineer (ACM, DCS, RC...)	► Architect (ARC)
► Competence Center Engineering Manager (CC EM)	► Design Authority (DA)
► Head of Discipline (HoD)	► Head of Engineering Delivery (HoED)
► Head of Engineering Operations (HoEO)	► Integration Verification Validation and Qualificat...
► Product Line Architect (PLA)	► System Engineer (SE)

Lifecycle

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In regard with project's constraints and solution complexity select the development life cycle among followings:

■ Orient
■ Design
■ Develop
■ Integrate & Verifiy
■ Validate



See 87201652-DDQ-GRP-EN Development Life Cycle guide

Tailoring

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a) Objective

DDQS tailoring aims to limit (tailoring down) or emphasize (tailoring up) engineering activities and practices to be performed to best fit with project and entity objectives taking into account the context (e.g. new product development, customer solution

development) and the solution complexity.

b) Principles

The engineering activities, reviews, roles, work products, practices and tools are tailored in accordance with the corporate DDQS tailoring instruction 87201604 and that of the entity if it exists. The resulting tailored process and practices with the rationale are recorded in the concerned engineering plan.

For a new product development, or a customer solution reusing product(s), the specific activities to perform and their interactions are detailed in the instruction referenced below.

Product Line Engineering (PLE) instruction, 87210086

Performance criteria

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Key Performance Indicators

The applicable performance criteria for this process are the Key Performance Indicators (KPI) listed in the "Definition of Indicators" instruction 87201738-QCI-GRP.

Other KPI can be added at Entity level. They shall be defined in the Measurement Plan.

Log of changes Group (inherited from: *Design, Develop and Qualify the Solution*)

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RELEASE	DESCRIPTION	DATE
L.2.2.0	CR080339: engineering delivery roles update - Architects, PDA, DA - EDM, EWPM (new role), IVVQM, SysEM, SwEM, HwEM - HoD, HoED (new role)	28/11/2023
L.2.1.0	CR077332: The Solution Engineering Manager (SEM) is renamed into Engineering Delivery Manager (EDM). No mission evolution.	27/06/2023
K.2.1.0	CR074796: due to the Engineering Make or Buy & Footprint guide publication update of the Manage Engineering Discipline activity description. New output on the activity: Make or Buy & Footprint (MoB&F) analysis and related transfer plan" New role : the MoB&F Manager has been added as contributor in several activities of DDQ (sol,sw,hw)	28/06/2022
K.2.0.1	CR073325: Due to the creation of 2.4 - S&OP process, creation of the Head of Engineering Operations (HoEO) and the Competence Center Engineering (CC EM) -> HoEO as contributor at DDQS top level process (Hw, Sol, Sw) -> HoEO as responsible for SOR review and for Manage Engineering Discipline (Hw; Sol, Sw)	05/04/2022
K.2.0.0	CR071840: Update PDA and SEM on the fact that PDA is focused on solution compliance and SEM is focused on delivery management	22/02/2022
J.2.2.0	CR070463: Engineering Maturity Leader (EML) renamed to Engineering Quality Leader (EQL) on Manage Engineering Discipline activity	26/10/2021
I.2.2.0	CR059955: addition of the Airworthiness Certification Engineer (ACM, DCS, RCE) as contributor on Orient, Design, Develop, Integrate & Verify and Validate the Solution activities	27/10/2020
I.2.0.0	CR056423: minor corrections on SEM and HoD role missions	28/01/2020
H.2.2.0	CR057296: In input data of the activity: replace "Validated technology by a Main Gate review" by "List of concepts/technologies validated by a main gate" CR056934: PDA mission update: addition of the of: - Support capture, bid and project team to understand the context and customer needs. - Assess alternative solutions.	24/09/2019
H.2.0.0	CR053440: Update of DA, PDA and SEM roles. CR052425: The DAPL has been replaced by the Product Line Architect (PLA). Job description remains identical H.2.1.0 : Addition of missing change log in H.2.0.0 about Engineering documents management (0115) rule concerning the involvement of SEM on DVa (CR052482)	29/01/2019
G.2.0.3	CR050676: Addition of a new rule for be compliant with General Data Protection Regulation	29/05/2018
G.2.0.0	CR046985: Update of the rule 0115: Engineering documents management (Solution) in consistency with RT04	31/01/2018
F.2.0.5	CR 46481: Renumbering of activity rules on 4 digits in accordance with the instruction for data management of the Management System.	27/09/2017
F.2.0.0	CR036118: General update to emphasize product engineering as solution engineering. Review of attached documents (remove of templates and guides transferred to TRS). Change of the map with addition of Manage engineering discipline. Update of main interprocess inputs/outputs. Creation of the activity "Manage Engineering Discipline" following the publication of 87210749-	29/03/2017

	GOV-GRP-EN Engineering Governance Instruction.	
E.2.0.2	CR035438 : Removing of Engineering Environment instruction	30/03/2016
E.2.0.0	CR030616 :Attachment of the new PLE instruction CR006836 : Update of key performance indicators CR 31359: Removal of rule 0) Enterprise Risk mitigation CR033062 : Addition of Inputs/Outputs data at the top level of the process following a major	27/01/2016
D.2.0.3	CR020803: Transfert of the PLE Handbook from DDQS TRS to attach it at DDQS Sol/Sw/Hw process top level in the PRS	27/05/2015
C.2.0.5	Evolution for simplification and the introduction of incremental/evolutive development cycles.	24/09/2014
C.2.0.0	-Clean up-Co-engineering taken in to account in Rule b)	29/01/2014
B.2.0.9	C00007637 - Move of "Manage Enterprise Risks with Chorus 2.0" instruction to Group TRS	29/07/2013
B.2.0.6	- Addition of Agile and Engineering Environment Instructions- Addition of the French version "Development Life Cycle Guide"- "Technical Standards" document has been removed- "Complexity Profiler Instruction" has been removed	26/04/2013
B.2.0.3	CIICR-1028 : Risks instruction and rules 0) added	30/01/2013
B.2.0.0	Cleanup, correction of consistency issues (CIICR-783)	26/09/2012
A.2.0.7	The D&JF French Instruction has been published (CIICR-607)	25/01/2012
A.2.0	Update CMMI (CIICR-301), Interface update (CIICR-346)	07/06/2011
A.1.0	CIICR-90, CIICR- 216, CIICR-219	15/04/2011
1.0	CHORUS 2.0 Go-live	10/03/2011

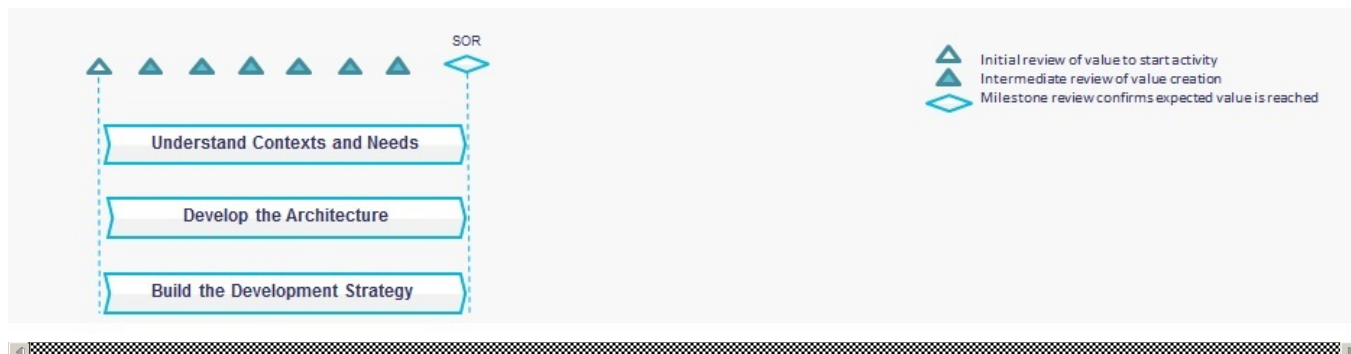
Note: Subject of the changes of the documents are detailed in the [Release History page](#).

RESPONSIBLE	
ACTORS	NAME AND ROLE
Content responsible	
Quality responsible	

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1 - Orient (ref: 87201340-00)

Activity flow diagram



Presentation

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Objective

To converge towards a balanced value proposition, building collectively a mutually agreed vision of:

- the solution to provision for a bid or a product,
- the processes sustaining the whole life cycle of the solution.

Description

These upstream activities are performed on the scope of a solution or solution element in co-engineering with related sub-elements or components activities. They aim at reaching a clear understanding of contexts and needs and providing the key elements about architecture and development strategy of that given solution or solution element.

Documents

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TITLE / REFERENCE	TYPE	SCOPE	SPECIFIC	LANGUAGE	DATE
CheckList for Orienting EcoDesign (CLOE) 87218708-DDQ-GRP-EN-001 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=271546551	TRS	Template and Form	Hardware Solution	EN	21/04/2023
Cybersecurity Engineering Policy Writing Guide 87213428-DDQ-GRP-EN-005 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=166409341	TRS	Guideline	Solution	EN	27/06/2023
Design Validation Form (DVa) 87201634-DDQ-GRP-EN-011 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=119176090	TRS	Template and Form	Solution	EN	27/06/2023
PBS - Product Breakdown Structure 87208986-DDQ-GRP-EN-006 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=109677034	TRS	Template and Form	Hardware Software Solution	EN	15/12/2021
Preferred Part List (PPL-PML) 87201845-DDQ-GRP-EN-003 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=276928511&vernum=1		Guideline	Hardware	EN	25/07/2023
Security engineering instruction 87210206-DDQ-GRP-EN-005 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=96019973&vernum=11		Instruction		EN	27/06/2023
Training Material-DASOR.pptx 65734404-DDQ-DIF-EN-002 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=181425813	TRS	Training	DIS	EN	22/07/2019
UX Design in Engineering 87216348-DDQ-GRP-EN-001 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=218307077&vernum=3		Instruction	Solution	EN	28/09/2021

Responsible

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► Responsible

► Contributor

► MoB&F Manager

Log of changes Group (inherited from: *Orient*)[Back to activity](#)

RELEASE	DESCRIPTION	DATE
I.2.2.0	CR059955: addition of the Airworthiness Certification Engineer (ACM, DCS, RCE) as contributor to all activities of Orient the Solution	27/10/2020
H.2.2.0	CR057296: In input data of the activity Orient/Understand contexts and needs: ■ replace "Validated technology by a Main Gate review" by " List of concepts/technologies validated by a main gate"	24/09/2019
F.2.0.5	CR046162 Replacement of the complete reference of the Engineering reviews instruction by the root of the document in the reviews description CR041811: Addition of Speciality engineer and DAPL roles to the Orient activities	27/09/2017
F.2.0.0	CR036118 full rework on structure and content to promote a full co-engineering across disciplines Sys/Sw/Hw.	29/03/2017
E.2.0.2	CR 35056 : Correction of activity names (not unique)	30/03/2016
E.2.0.0	CR030615:Update for installation for Understand needs activity	27/01/2016
D.2.0.3	CR022038: Removal of obsolete document: 61497416-SER-GRP-EN Guide for Preparing the logistics Part of Proposals CR022054: To attach the new installation guide to Activity : understand needs	27/05/2015
D.2.0.0	CR007043: Update for Product Line Engineering CR007727: new template PBS attached "to build the solution strategy"	28/01/2015
C.2.0.5	CR5149 :- precisions for incremental/evolutive development cycles- replacement of engineering disciplines role by System Engineer (SE)	24/09/2014
C.2.0.0	- Minor corrections (spelling, correction or addition of Inputs/Outputs, layout...)- Attachment of co-engineering instruction and COTS & OSS instruction- Attachment of a new instruction "Instruction for Establishing a Product-PMS Technical Desc	29/01/2014
B.2.0.6	Addition of the French version « Development Life Cycle Guide »	26/04/2013
B.2.0.0	Cleanup, correction of coconsistency issues (CIICR-783), Evolution inputs/outputs/role & attached documents (CIICR-409)	26/09/2012
A.2.0.7	Complexity Profiler has been added (CIICR-553), The D&JF French Instruction has been published (CIICR-607)	25/01/2012
A.2.0.3	Complexity Profiler add-on (CIICR-431)	29/09/2011
A.2.0	Interface update (CIICR-346)	07/06/2011
A.1.0	CIICR-90, CIICR- 216, CIICR-219	15/04/2011
1.0	CHORUS 2.0 Go-live	10/03/2011

Note: Subject of the changes of the documents are detailed in the [Release History page](#).

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1.1 - Understand Contexts and Needs (ref: 87201340-01)

Activity flow diagram

Presentation

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Objective

To embrace the various contexts in all their extent and understand what is at stakes.

Description

Identify all concerned stakeholders both internal and external.

Involve identified stakeholders understanding their stakes and needs.

Embrace with a broad approach the capture of information and knowledge about the various concerned contexts.

Synthesise stakeholders' key expectations or needs, along with major constraints driving the shaping of the solution.

Documents

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TITLE / REFERENCE	TYPE	SCOPE	SPECIFIC	LANGUAGE	DATE
HUMAN_SYSTEMS_INTEGRATION_GUIDE 87203104-DDQ-GRP-EN-004 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=19447145	Guide line			EN	16/09/2022
Instruction for producing Design & Justification File (D&JF) 87201508-DDQ-GRP-EN-009 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=96091671&vern=12	Instruction		Hardware Software Solution	EN	28/11/2023 U!
OCD template 87210848-DDQ-GRP-EN-006 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=85758611	Template and Form		Solution	EN	27/06/2023
Solution Engineering with Agile 87211549-DDQ-GRP-EN-003 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=128280233	Guide line			EN	27/06/2023

Inputs/Outputs

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Inputs

Current technological and production constraints
 Customer legacy and target location data
 Customer requirements, requests and needs
 Internal Executive summary - BKO
 List of concepts/technologies validated by a main gate
 Market & Customer Requirement Description (for SOR)
 Product Line strategy with roadmap
 QHSE policy applicable to the entity
 Validated SBP
 Value proposition, Business model and Pricing guideline

Outputs

Contexts Description (OCD)
 Shared Vision of the Goal (OCD)

Responsible

[Back to activity](#)

► Responsible ► Contributor

► Project Design Authority (PDA)	► Airworthiness Certification Engineer (ACM, DCS, RC...
► Architect (ARC)	► Design Authority (DA)
► Engineering Delivery Manager (EDM)	► Hardware Architect (HwARC)
► Hardware Engineering Manager (HwEM)	► Product Line Architect (PLA)
► Product Line Manager (PLM)	► Software Architect (SwARC)

▶ Software Engineering Manager (SwEM)	▶ System Engineer (SE)
▶ System Engineering Manager (SysEM)	

1.2 - Develop the Architecture (ref: 87201340-04)

Activity flow diagram

Presentation

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Objective

To build collectively a viable architecture.

Description

Bring to the foreground the key architectural drivers.

Identify the key architectural decisions.

Explore the solution space envisaging a set of alternate architectural concepts and perform related trade-offs.

Perform early validation of the architecture.

Describe, share and promote the architecture.

Documents

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TITLE / REFERENCE	TYPE	SCOPE	SPECIFIC	LANGUAGE	DATE
A3AO guidelines 87209446-DDQ-GRP-EN-003 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=59737389	TRS	Guide line		EN	29/09/2020
ARCADIA reference manual 87201608-DDQ-GRP-EN-002 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=19443354	TRS	Guide line	Solution	EN	05/09/2012
Cyber Security guide for solution architects 87210649-DDQ-GRP-EN-004 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=85760869	TRS	Guide line	Solution	EN	11/05/2022
Cybersecurity Guide for Equipment Architects 87214870-DDQ-GRP-EN-004 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=191297848	TRS	Guide line	Solution	EN	29/06/2022
Cybersecurity Risk Management Guide 87210648-DDQ-GRP-EN-004 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=85754580	TRS	Guide line		EN	27/06/2023
Design Validation Form (DVa) 87201634-DDQ-GRP-EN-011 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=119176090	TRS	Template and Form	Solution	EN	27/06/2023
GUIDE FOR SOFTWARE COTS AND OPEN SOURCE MANAGEMENT 61565035-DDQ-GRP-EN-005 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=19487973	TRS	Guide line	Software	EN	29/06/2021
Guide for Co-Engineering practice implementation 87206365-DDQ-GRP-EN-004 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=41267858	TRS	Guide line		EN	27/06/2023
Guide for Concept Development & Experimentation 87208105-DDQ-GRP-EN-003 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=46601147	TRS	Guide line	Solution	EN	27/06/2023
HUMAN_SYSTEMS_INTEGRATION_GUIDE 87203104-DDQ-GRP-EN-004 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=19447145	TRS	Guide line		EN	16/09/2022

TITLE / REFERENCE	TYPE	SCOPE	SPECIFIC	LANGU AGE	DATE
Instruction for producing Design & Justification File (D&JF) 87201508-DDQ-GRP-EN-009 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=96091671&vernum=12	Instru ction		Hard ware Softw are Soluti on	EN	28/11/2023 <i>U!</i>
OCD template 87210848-DDQ-GRP-EN-006 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=85758611	TRS Templ ate and Form		Soluti on	EN	27/06/2023
SYSTEM ARCHITECTING GUIDE 87202704-DDQ-GRP-EN-002 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=19446147	TRS Guide line		Soluti on	EN	27/06/2023
Safety_Engineering_Guidance 87206928-DDQ-GRP-EN-001 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=30372497	TRS Guide line			EN	22/06/2013
Solution Engineering with Agile 87211549-DDQ-GRP-EN-003 =128280233	TRS Guide line			EN	27/06/2023

Inputs/Outputs

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Inputs

Contexts Description (OCD)
Current technological and production constraints
Customer legacy and target location data
Customer requirements, requests and needs
Product Line strategy with roadmap
QHSE policy applicable to the entity
Shared Vision of the Goal (OCD)
Validated SBP

Outputs

Architecture Description (D&JF)
Architecture Justification (D&JF)

Responsible

[Back to activity](#)

► Responsible ► Contributor

► Project Design Authority (PDA)	► Airworthiness Certification Engineer (ACM, DCS, RC...)
► Architect (ARC)	► Design Authority (DA)
► Engineering Delivery Manager (EDM)	► Hardware Architect (HwARC)
► Hardware Engineering Manager (HwEM)	► MoB&F Manager
► Procurement Product/Bid/Project Manager (PPM)	► Product Line Architect (PLA)
► Software Architect (SwARC)	► Software Engineering Manager (SwEM)
► Speciality Engineer (SPE)	► System Engineer (SE)
► System Engineering Manager (SysEM)	

1.3 - Build the Development Strategy (ref: 87201340-02)

Activity flow diagram

Presentation

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Objective

To elaborate the strategy governing the development activities to reach the solution as described in the architecture, fostering cooperation, between Engineering, Industry and Services.

Description

Define the Top Level PBS

For the solution, and recursively lower level elements:

- Define, plan and synchronize the various elements' lifecycles.
- Build IVVQ strategy and reconcile with other relevant strategies.
- Select development lifecycles and approaches.
- Plan integration of required engineering disciplines and specialties.

Documents

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TITLE / REFERENCE	TYPE	SCOPE	SPECIFIC	LANGUAGE	DATE
Automated HCI Testing Guidance 87209386-DDQ-GRP-EN-001 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=59728158	TRS	Guide line	Software	EN	13/01/2015
Complexity Profiler Guide 87203444-DDQ-GRP-EN-007 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=88235717	TRS	Guide line	Solution	EN	27/06/2023
Complexity_Profiler template 87203605-DDQ-GRP-EN-004 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=88263544	TRS	Template and Form	Solution	EN	26/05/2020
Cybersecurity Engineering Guide 87210647-DDQ-GRP-EN_005 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=85758845	TRS	Guide line		EN	27/06/2023
Cybersecurity Guide for Equipment Architects 87214870-DDQ-GRP-EN-004 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=191297848	TRS	Guide line	Solution	EN	29/06/2022
Cybersecurity Risk Management Guide 87210648-DDQ-GRP-EN-004 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=85754580	TRS	Guide line		EN	27/06/2023
DIVVA Core Model Description for IVVQ Management 87211648-DDQ-GRP-EN-005 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=119867105	TRS	Guide line		EN	16/09/2022
Development Life Cycle Guide 87201652-DDQ-GRP-EN-011 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=109677175	TRS	Guide line	Solution	EN	27/06/2023
Engineering Management Plan (EMP) writing guide 87216448-DDQ-GRP-EN-006 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=215704008	TRS	Guide line	Hardware Software Solution	EN	27/06/2023
GUIDE FOR SOFTWARE COTS AND OPEN SOURCE MANAGEMENT 61565035-DDQ-GRP-EN-005	TRS	Guide line	Software	EN	29/06/2021

TITLE / REFERENCE	TYPE	SCOPE	SPECIFIC	LANGU AGE	DATE
https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=19487973					
TRS Guide for Co-Engineering practice implementation 87206365-DDQ-GRP-EN-004 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=41267858	Guide line			EN	27/06/2023
TRS Guide for vulnerability assessment during IVVQ activities 87215508-DDQ-GRP-EN-001 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=200149122	Guide line		Soluti on	EN	28/07/2020
TRS HUMAN_SYSTEMS_INTEGRATION_GUIDE 87203104-DDQ-GRP-EN-004 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=19447145	Guide line			EN	16/09/2022
TRS IVVQ Handbook 87100124-DDQ-GRP-EN-009 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=19448365	Guide line			EN	27/06/2023
Instruction for producing Design & Justification File (D&JF) 87201508-DDQ-GRP-EN-009 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=96091671&vernum=12	Instru ction		Hard ware Softw are Soluti on	EN	28/11/2023 U!
TRS Manage the product Policy tailoring guidelines for Agile Product Management 87217208-PRP-GRP-EN-001 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=223836835	Guide line			EN	05/10/2021
TRS PBS - Product Breakdown Structure 87208986-DDQ-GRP-EN-006 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=109677034	Templ ate and Form		Hard ware Softw are Soluti on	EN	15/12/2021
TRS PP Toolbox 07-Technical & Industrial Synthesis 2017 87212096-PRP-GRP-EN-002 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=131389604	Guide line			EN	29/01/2019
TRS Practice Building Block key Requirements 87212104-PRP-GRP-EN-002 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=131390351	Guide line			EN	29/01/2019
TRS Practice Dream Product Competitiveness Assessment Grid 87212101-PRP-GRP-EN-001 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=131392295	Guide line			EN	27/09/2017
TRS Practice PLE Product Reuse Strategy decision support 87212106-PRP-GRP-EN-002 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=131389602	Guide line			EN	29/01/2019
TRS Practice Product_PL Roadmap Template 87212105-PRP-GRP-EN-001 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=131393880	Guide line			EN	27/09/2017
TRS Product_Line_Engineering_PLE_Handbook 87208865-DDQ-GRP-EN-003 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=109677387	Guide line		Hard ware Softw are Soluti on	EN	27/06/2023
TRS Safety_Engineering_Guidance 87206928-DDQ-GRP-EN-001 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=30372497	Guide line			EN	22/06/2013
TRS Solution Engineering with Agile 87211549-DDQ-GRP-EN-003 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=128280233	Guide line			EN	27/06/2023

Inputs/Outputs

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Inputs

Architecture Description (D&JF)
 Architecture Justification (D&JF)
 Contexts Description (OCD)
 Customer requirements, requests and needs
 List of concepts/technologies validated by a main gate
 Shared Vision of the Goal (OCD)

Outputs

IVVQ Strategy
 Integrated Solution Development Strategy
 Solution validation (DVa)
 Top level PBS

Responsible

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► Responsible ► Contributor

► Engineering Delivery Manager (EDM)	► Airworthiness Certification Engineer (ACM, DCS, RC...
► Architect (ARC)	► Configuration Manager
► Design Authority (DA)	► Hardware Architect (HwARC)
► Hardware Engineering Manager (HwEM)	► Integration Verification Validation and Qualificat...
► MoB&F Manager	► Procurement Product/Bid/Project Manager (PPM)
► Product Line Architect (PLA)	► Project Design Authority (PDA)
► Software Architect (SwARC)	► Software Engineering Manager (SwEM)
► System Engineer (SE)	► System Engineering Manager (SysEM)

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1.4 - SOR (Solution Orientation Review) (ref: 87202127-01)

Activity flow diagram



Presentation

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Objective

To approve the target architecture and the development strategy achieving a value proposition with regard to risks, costs and satisfaction of the stakeholders.

Description

This review is performed on the scope of the solution or solution element with full involvement of the hardware and software disciplines.

For details see 87211053 Engineering reviews instruction.

Documents

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TITLE / REFERENCE		TYPE	SCOPE	SPECIFIC	LANGUAGE	DATE
DDQS reviews checklists 87211132-DDQ-GRP-EN-018	TRS	Template and Form		Hardware Software Solution	EN	27/06/2023
https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=120003684						
Engineering reviews instruction 87211053-DDQ-GRP-EN-006		Instruction			EN	11/07/2023
https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=119567360&vernum=9						
Export control Checklist 87215152-DDQ-GRP-EN-003	TRS	Template and Form		Hardware Software Solution	EN	27/06/2023
https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=200146792						

Responsible

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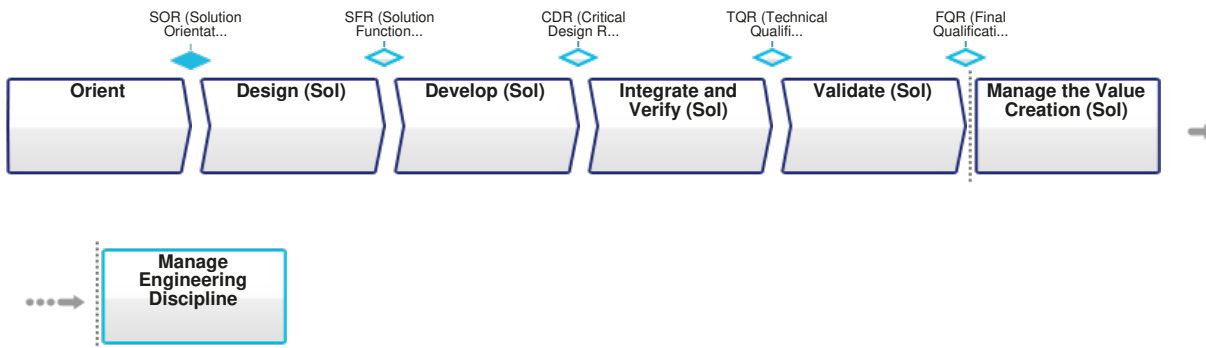
► Responsible ► Contributor

► Competence Center Engineering Manager (CC EM)	► Head of Discipline (HoD)
► Head of Engineering Delivery (HoED)	► Head of Engineering Operations (HoEO)
► Project Design Authority (PDA)	► Airworthiness Certification Engineer (ACM, DCS, RC...)
► Engineering Delivery Manager (EDM)	► Hardware Architect (HwARC)
► Hardware Engineering Manager (HwEM)	► Integration Verification Validation and Qualificat...
► MoB&F Manager	► Procurement Product/Bid/Project Manager (PPM)
► Production Process & Technology Manager (PPTM)	► Quality Assurance Manager (QAM)
► Service Engineering Manager (SvEM)	► Software Architect (SwARC)
► Software Engineering Manager (SwEM)	► System Engineering Manager (SysEM)

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2 - SOR (Solution Orientation Review) (ref: 87202127-01)

Activity flow diagram



Presentation

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Objective

To approve the target architecture and the development strategy achieving a value proposition with regard to risks, costs and satisfaction of the stakeholders.

Description

This review is performed on the scope of the solution or solution element with full involvement of the hardware and software disciplines.

For details see 87211053 Engineering reviews instruction.

Documents

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TITLE / REFERENCE	TYPE	SCOPE	SPECIFIC	LANGUAGE	DATE
DDQS reviews checklists 87211132-DDQ-GRP-EN-018 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=120003684	TRS	Template and Form	Hardware Software Solution	EN	27/06/2023
Engineering reviews instruction 87211053-DDQ-GRP-EN-006 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=119567360&vernum=9	Instruction			EN	11/07/2023
Export control Checklist 87215152-DDQ-GRP-EN-003 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=200146792	TRS	Template and Form	Hardware Software Solution	EN	27/06/2023

Responsible

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► Responsible ► Contributor

► Competence Center Engineering Manager (CC EM)	► Head of Discipline (HoD)
► Head of Engineering Delivery (HoED)	► Head of Engineering Operations (HoEO)
► Project Design Authority (PDA)	► Airworthiness Certification Engineer (ACM, DCS, RC...)
► Engineering Delivery Manager (EDM)	► Hardware Architect (HwARC)
► Hardware Engineering Manager (HwEM)	► Integration Verification Validation and Qualificat...
► MoB&F Manager	► Procurement Product/Bid/Project Manager (PPM)
► Production Process & Technology Manager (PPTM)	► Quality Assurance Manager (QAM)

▶ Service Engineering Manager (SvEM)

▶ Software Engineering Manager (SwEM)

▶ Software Architect (SwARC)

▶ System Engineering Manager (SysEM)

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3 - Design (Sol) (ref: 87201334-00)

Activity flow diagram



Presentation

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Objective

To formalise any outstanding requirements, finalise and agree on the specification, the chosen design used to develop the solution.

Description

The above activities may be considered as sequenced or concurrent depending on the chosen development cycle.

Documents

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TITLE / REFERENCE	TYPE	SCOPE	SPECIFIC	LANG UAGE	DATE
DCD template 87213132-DDQ-GRP-EN-005 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=160797377	TRS	Template and Form	Hardware Software Solution	EN	27/06/2023
Design FMEA Template 65734410-DDQ-DIF-EN-002 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=231747238&vernum=2	Template and Form	DIS		EN	25/04/2023
Engineering Design documents writing guide 87216830-DDQ-GRP-EN-002 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=226820662	TRS	Guideline	Hardware Software Solution	EN	27/06/2023
Engineering specification documents writing guide 87216468-DDQ-GRP-EN-004 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=218749673	TRS	Guideline	Hardware Software Solution	EN	27/06/2023
FMEA Thales DIS Procedure 65734408-DDQ-DIF-EN-002 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=231751588&vernum=3	Instruction	DIS		EN	11/05/2023
Management & Use of external software components (COTS & OSS) 87202149-DDQ-GRP-EN-009 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=96091491&vernum=12	Instruction			EN	27/06/2023
Process FMEA Template 65734409-DDQ-DIF-EN-002 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=231752472&vernum=2	Template and Form	DIS		EN	25/04/2023

TITLE / REFERENCE	TYPE	SCOPE	SPECIFIC	LANG UAGE	DATE
Security engineering instruction 87210206-DDQ-GRP-EN-005 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=96019973&vernum=11	Instru ction			EN	27/06/2023
UX Design in Engineering 87216348-DDQ-GRP-EN-001 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=218307077&vernum=3	Instru ction		Soluti on	EN	28/09/2021

Responsible

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► Responsible ► Contributor

► MoB&F Manager

Log of changes Group (inherited from: *Design (Sol)*)[Back to activity](#)

RELEASE	DESCRIPTION	DATE
I.2.2.0	CR059955: addition of the Airworthiness Certification Engineer (ACM, DCS, RCE) as contributor to all activities of Design the Solution	27/10/2020
H.2.0.0	CR05125: Modification of Formalise requirements (Sol) description to introduce the notion of "special" requirement	29/01/2019
F.2.0.5	CR046162 Replacement of the complete reference of the Engineering reviews instruction by the root of the document in the reviews description	27/09/2017
F.2.0.0	CR036118: Renaming of « Build the architectural design » in « Design the solution » Rewording of descriptions to converge as far as possible between Sol/Sw/Hw.	29/03/2017
E.2.0.2	CR 35056 : Correction of activity names (not unique)	30/03/2016
E.2.0.0	CR030615 :Update for installation	27/01/2016
C.2.0.5	CR5149 :- precisions for incremental/evolutive development cycles- replacement of engineering disciplines role by System Engineer (SE)	24/09/2014
C.2.0.0	Minor corrections (spelling, correction or addition of Inputs/Outputs, layout...)	29/01/2014
B.2.0.0	Cleanup, correction of consistency issues (CIICR-783), Evolution inputs/outputs/role & attached documents (CIICR-409)	26/09/2012
A.2.0.7	The D&JF French Instruction has been published (CIICR-607)	25/01/2012
A.2.0	Update CMMI (CIICR-301)	19/05/2011
A.1.0	CIICR-90, CIICR- 216, CIICR-219	15/04/2011
1.0	CHORUS 2.0 Go-live	10/03/2011

Note: Subject of the changes of the documents are detailed in the [Release History page](#).

RESPONSIBLE	
ACTORS	NAME AND ROLE
Content responsible	
Quality responsible	

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3.1 - Formalise requirements (Sol) (ref: 87201334-01)

Activity flow diagram

Presentation

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Objective

Describe the selected solution in terms of capabilities, characterized by a set of stakeholder functional and non functional requirements, which allows the establishment of the Functional Baseline

Description

Analyse, prioritise and validate stakeholders' requirements.

Identifying special requirements.

Formalise the specification and the interface of the solution or solution element.

Establish and maintain the Functional Baseline (FBL)

Documents

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TITLE / REFERENCE	TYPE	SCOPE	SPECIFIC	LANGUAGE	DATE
Engineering specification documents writing guide 87216468-DDQ-GRP-EN-004 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=218749673	Guide line		Hard ware Softw are Soluti on	EN	27/06/2023
HUMAN_SYSTEMS_INTEGRATION_GUIDE 87203104-DDQ-GRP-EN-004 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=19447145	Guide line			EN	16/09/2022
OCD template 87210848-DDQ-GRP-EN-006 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=85758611	Templ ate and Form		Soluti on	EN	27/06/2023

Inputs/Outputs

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Inputs

Architecture Description (D&JF)

Architecture Justification (D&JF)

Contexts Description (OCD)

SOR Report

Service Definition Document

Shared Vision of the Goal (OCD)

Outputs

External interfaces specifications (ICD)

Functional Baseline (FBL)

Specifications (SSS / PIDS)

Responsible

[Back to activity](#)

► Responsible ► Contributor

► Engineering Delivery Manager (EDM)	► Airworthiness Certification Engineer (ACM, DCS, RC...
► Architect (ARC)	► Configuration Manager
► Integration Verification Validation and Qualificat...	► Production Process & Technology Manager (PPTM)
► Service Engineering Manager (SvEM)	► Speciality Engineer (SPE)
► System Engineer (SE)	► System Engineering Manager (SysEM)

[Table of contents](#) > [Design, Develop and Qualify the Solution](#) > [Design \(Sol\)](#) > **SRR (Solution Requirement Review)**

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3.2 - SRR (Solution Requirement Review) (ref: 87201334-02)

Activity flow diagram



Presentation

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Objective

Verify that all solution requirements have been properly identified and that customer and supplier understand each other position and commitments.

Description

For details see 87211053 Engineering reviews instruction.

Documents

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TITLE / REFERENCE	TYPE	SCOPE	SPECIFIC	LANGUAGE	DATE
DDQS reviews checklists 87211132-DDQ-GRP-EN-018 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=120003684	TRS	Template and Form	Hardware Software Solution	EN	27/06/2023
Engineering reviews instruction 87211053-DDQ-GRP-EN-006 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=119567360&vernum=9	Instruction			EN	11/07/2023
Export control Checklist 87215152-DDQ-GRP-EN-003 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=200146792	TRS	Template and Form	Hardware Software Solution	EN	27/06/2023

Responsible

[Back to activity](#)

► Responsible ► Contributor

► Project Design Authority (PDA)	► Airworthiness Certification Engineer (ACM, DCS, RC...
► Engineering Delivery Manager (EDM)	► Integration Verification Validation and Qualificat...
► Production Process & Technology Manager (PPTM)	► Quality Assurance Manager (QAM)
► Service Engineering Manager (SvEM)	► System Engineering Manager (SysEM)

3.3 - Design the solution (ref: 87201334-03)

Activity flow diagram

Presentation

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Objective

To construct the optimal design, which can be developed and delivered to satisfy the customer's and other stakeholders' needs.

Description

Perform the design choice using criteria.
 Develop the system logical architecture.
 Create the system physical architecture by developing and analysing alternatives taking into consideration constraints (i.e. legacy, product line strategy, deployment strategy, etc).
 Define internal interfaces accordingly with product breakdown and Integration / Verification strategy.
 Refine and allocate the solution's or solution element's requirements to the lower level elements.
 Provide consistent interface specifications.
 Formalise the design of the solution or solution element.
 Development of the preliminary specifications of the lower level elements.

Documents

[Back to activity](#)

TITLE / REFERENCE	TYPE	SCOPE	SPECIFIC	LANGUAGE	DATE
Cyber Security guide for solution architects 87210649-DDQ-GRP-EN-004 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=85760869	TRS	Guide line	Solution	EN	11/05/2022
Cybersecurity Engineering Guide 87210647-DDQ-GRP-EN-005 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=85758845	TRS	Guide line		EN	27/06/2023
Cybersecurity Engineering Policy Writing Guide 87213428-DDQ-GRP-EN-005 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=166409341	TRS	Guide line	Solution	EN	27/06/2023
Cybersecurity Risk Management Guide 87210648-DDQ-GRP-EN-004 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=85754580	TRS	Guide line		EN	27/06/2023
DCD template 87213132-DDQ-GRP-EN-005 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=160797377	TRS	Template and Form	Hardware Software Solution	EN	27/06/2023
Engineering Design documents writing guide 87216830-DDQ-GRP-EN-002 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=226820662	TRS	Guide line	Hardware Software Solution	EN	27/06/2023
Feature Model Template 87209707-DDQ-GRP-EN-006 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=65640486	TRS	Template and Form		EN	27/06/2023
Feature Modelling Guide 87209145-DDQ-GRP-EN-001 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=57144308	TRS	Guide line		EN	01/12/2014

TITLE / REFERENCE	TYPE	SCOPE	SPECIFIC	LANGUAGE	DATE
Guide for Concept Development & Experimentation 87208105-DDQ-GRP-EN-003 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=46601147	Guide line		Solution	EN	27/06/2023
HUMS engineering guide 87209846-DDQ-GRP-EN-004 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=82756462	Guide line		Hardware Software Solution	EN	27/06/2023
Instruction for producing Design & Justification File (D&JF) 87201508-DDQ-GRP-EN-009 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=96091671&vernum=12	Instruction		Hardware Software Solution	EN	28/11/2023 U!
SYSTEM ARCHITECTING GUIDE 87202704-DDQ-GRP-EN-002 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=19446147	Guide line		Solution	EN	27/06/2023

Inputs/Outputs

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Inputs

Architecture Description (D&JF)
Architecture Justification (D&JF)
Contexts Description (OCD)
External interfaces specifications (ICD)
IVVQ Strategy
Shared Vision of the Goal (OCD)
Site surveys' and legacy's detailed data
Specifications (SSS / PIDS)

Outputs

Design & Justification File (D&JF)
Internal interfaces specifications (ICD)
Solution design document (SSDD)

Responsible

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► Responsible ► Contributor

► Engineering Delivery Manager (EDM)	► Airworthiness Certification Engineer (ACM, DCS, RC...)
► Architect (ARC)	► Configuration Manager
► Integration Verification Validation and Qualificat...	► MoB&F Manager
► Production Process & Technology Manager (PPTM)	► Service Engineering Manager (SvEM)
► Speciality Engineer (SPE)	► System Engineer (SE)
► System Engineering Manager (SysEM)	

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3.4 - SFR (Solution Functional Review) (ref: 87202127-02)

Activity flow diagram



Presentation

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Objective

To confirm that the design is compliant with the solution requirements.

Description

For details see 87211053 DDQS reviews instruction.

Documents

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TITLE / REFERENCE	TYPE	SCOPE	SPECIFIC	LANGUAGE	DATE
DDQS reviews checklists 87211132-DDQ-GRP-EN-018 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=120003684	TRS Template and Form		Hard ware Softw are Soluti on	EN	27/06/2023
Engineering reviews instruction 87211053-DDQ-GRP-EN-006 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=119567360&vernum=9	Instru ction			EN	11/07/2023
Export control Checklist 87215152-DDQ-GRP-EN-003 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=200146792	TRS Template and Form		Hard ware Softw are Soluti on	EN	27/06/2023

Responsible

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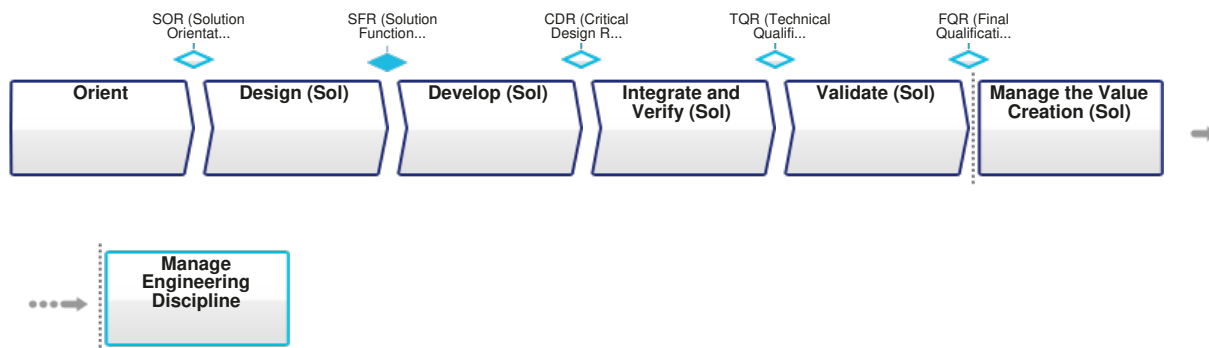
► Responsible ► Contributor

► Project Design Authority (PDA)	► Airworthiness Certification Engineer (ACM, DCS, RC...
► Engineering Delivery Manager (EDM)	► Integration Verification Validation and Qualificat...
► MoB&F Manager	► Production Process & Technology Manager (PPTM)
► Quality Assurance Manager (QAM)	► Service Engineering Manager (SvEM)
► System Engineering Manager (SysEM)	

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4 - SFR (Solution Functional Review) (ref: 87202127-02)

Activity flow diagram



Presentation

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Objective

To confirm that the design is compliant with the solution requirements.

Description

For details see 87211053 DDQS reviews instruction.

Documents

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TITLE / REFERENCE	TYPE	SCOPE	SPECIFIC	LANGUAGE	DATE
DDQS reviews checklists 87211132-DDQ-GRP-EN-018 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=120003684	TRS	Template and Form	Hard ware Softw are Soluti on	EN	27/06/2023
Engineering reviews instruction 87211053-DDQ-GRP-EN-006 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=119567360&vernum=9	Instru ction			EN	11/07/2023
Export control Checklist 87215152-DDQ-GRP-EN-003 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=200146792	TRS	Template and Form	Hard ware Softw are Soluti on	EN	27/06/2023

Responsible

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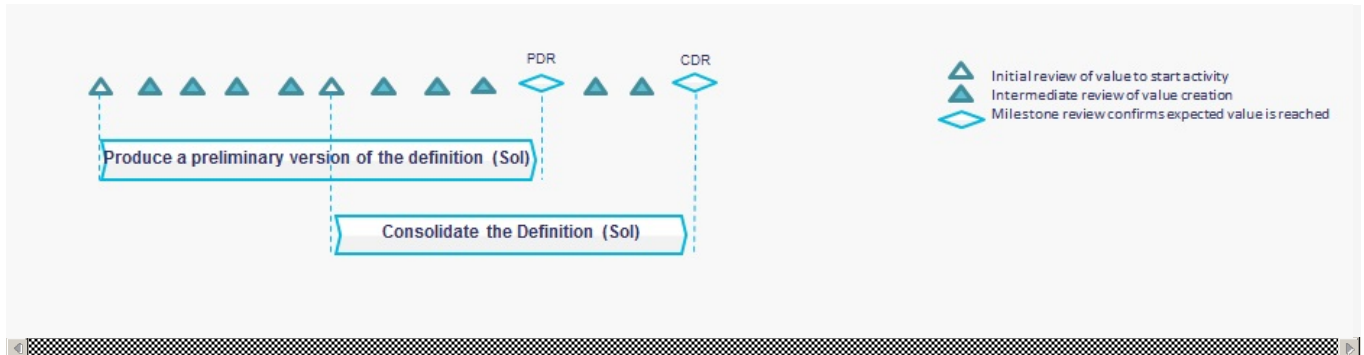
► Responsible ► Contributor

► Project Design Authority (PDA)	► Airworthiness Certification Engineer (ACM, DCS, RC...
► Engineering Delivery Manager (EDM)	► Integration Verification Validation and Qualificat...
► MoB&F Manager	► Production Process & Technology Manager (PPTM)
► Quality Assurance Manager (QAM)	► Service Engineering Manager (SvEM)
► System Engineering Manager (SysEM)	

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5 - Develop (Sol) (ref: 87201335-00)

Activity flow diagram



Presentation

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Objective

To refine the lower level of requirements down to a complete definition (Solution elements, HW and/or SW components) and to consolidate definition up to the solution level with agreement of all concerned levels.

Description

The above activities may be considered as sequenced, overlapped or concurrent depending on the chosen development cycle.

Documents

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TITLE / REFERENCE	TYPE	SCOPE	SPECIFIC	LANGUAGE	DATE
Cybersecurity Engineering Policy Writing Guide 87213428-DDQ-GRP-EN-005 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=166409341	TRS Guide line		Solution	EN	27/06/2023
Security engineering instruction 87210206-DDQ-GRP-EN-005 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=96019973&vernum=11	Instruction			EN	27/06/2023
UX Design in Engineering 87216348-DDQ-GRP-EN-001 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=218307077&vernum=3	Instruction		Solution	EN	28/09/2021

Roles

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No role for the selected object

Log of changes Group (inherited from: *Develop (Sol)*)

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RELEASE	DESCRIPTION	DATE
I.2.2.0	CR059955: addition of the Airworthiness Certification Engineer (ACM, DCS, RCE) as contributor for the all activities for Develop the Solution	27/10/2020
F.2.0.5	CR046162 Replacement of the complete reference of the Engineering reviews instruction by the root of the document in the reviews description	27/09/2017
F.2.0.0	CR036118: Renaming of « Finalise the solution definition » in « Finalise the solution » to be more generic. Elimination of useless wording.	29/03/2017
E.2.0.2	CR 35056 : Correction of activity names (not unique)	30/03/2016

E.2.0.0	CR29624:Correction of the objectives of the following activities : Produce a preliminary version of the definition and Finalise solution definition	27/01/2016
C.2.0.5	CR5149 :- precisions for incremental/evolutive development cycles- replacement of engineering disciplines role by System Engineer (SE) - Suppression of Monitor and control components implementation activity (redundancy with Monitor and control the development of Manage & support	24/09/2014
C.2.0.0	Minor corrections (spelling, correction or addition of Inputs/Outputs, layout...)	29/01/2014
B.2.0.0	Cleanup, correction of consistency issues (CIICR-783)	26/09/2012
A.2.0.7	The D&JF French Instruction has been published (CIICR-607)	25/01/2012
A.2.0	Interface update (CIICR-346)	07/06/2011
A.1.0	CIICR-90, CIICR- 216, CIICR-219	15/04/2011
1.0	CHORUS 2.0 Go-live	10/03/2011

Note: Subject of the changes of the documents are detailed in the [Release History page](#).

RESPONSIBLE	
ACTORS	NAME AND ROLE
Content responsible	
Quality responsible	

[Table of contents](#) > [Design, Develop and Qualify the Solution](#) > [Develop \(Sol\)](#) > **Produce a preliminary version of the definition (Sol)**

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5.1 - Produce a preliminary version of the definition (Sol) (ref: 87201335-01)

Activity flow diagram



Presentation

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Objective

To consolidate preliminary design of lower levels elements or components.

Description

Revisit the component functional requirements,
Adjust the component non-functional requirements,
Control the requirements specifications,
Revisit the interface needs and constraints,
Specify the interface between components,
Conduct component requirements validation,
Review the component requirements validation results,
Establish the initial Product Baseline (PBL).

Documents

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TITLE / REFERENCE	TYPE	SCOPE	SPECIFIC	LANGUAGE	DATE
Engineering specification documents writing guide 87216468-DDQ-GRP-EN-004	Guide line		Hard ware Softw are Soluti on	EN	27/06/2023
https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=218749673					

Inputs/Outputs

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Inputs

Allocated Baseline (ABL)
Requirements & architecture justification of solution elements (D&JF)
SFR Report
Solution design document (SSDD)

Outputs

Product data Package (PDP)
Requirements & architecture justification of solution elements (D&JF)
Solution elements' interface documents (ICD/IRS)
Solution elements' specifications (SSS/PIDS/SRS)

Responsible

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► Responsible ► Contributor

► Engineering Delivery Manager (EDM)	► Airworthiness Certification Engineer (ACM, DCS, RC...
► Architect (ARC)	► Configuration Manager
► Integration Verification Validation and Qualificat...	► System Engineer (SE)
► System Engineering Manager (SysEM)	

5.2 - PDR Preliminary Design Review for Solution (ref: 87201335-02)

Activity flow diagram



Presentation

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Objective

- Assess the preliminary design of the solution, including all its components items and interfaces,
- Assess the level of detail of the preliminary design
- Assess the compliance of the preliminary design with the solution requirements,
- Assess the completion of initial Product Baseline (PBL).

Description

For details see 87211053 Engineering Reviews instruction.

Documents

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TITLE / REFERENCE	TYPE	SCOPE	SPECIFIC	LANGUAGE	DATE
DDQS reviews checklists 87211132-DDQ-GRP-EN-018 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=120003684	TRS	Template and Form	Hardware Software Solution	EN	27/06/2023
Engineering reviews instruction 87211053-DDQ-GRP-EN-006 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=119567360&vernum=9		Instruction		EN	11/07/2023
Export control Checklist 87215152-DDQ-GRP-EN-003 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=200146792	TRS	Template and Form	Hardware Software Solution	EN	27/06/2023

Responsible

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► Responsible ► Contributor

► Project Design Authority (PDA)	► Airworthiness Certification Engineer (ACM, DCS, RC...)
► Engineering Delivery Manager (EDM)	► Integration Verification Validation and Qualificat...
► MoB&F Manager	► Production Process & Technology Manager (PPTM)
► Quality Assurance Manager (QAM)	► Service Engineering Manager (SvEM)
► System Engineering Manager (SysEM)	

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5.3 - Consolidate the definition (Sol) (ref: 87201335-03)

Activity flow diagram



Presentation

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Objective

To consolidate detailed design of lower levels elements or components.

Description

Detail the requirements specifications.

Finalise the interface definition (until detailed description and drawings).

Ensure the compliance between the initial Product Baseline (PBL) and the Allocated Baseline (ABL).

Establish the Product Baseline (PBL) and provide the Design & Justification File (D&JF)

Documents

[Back to activity](#)

TITLE / REFERENCE	TYPE	SCOPE	SPECIFIC	LANGUAGE	DATE
No document attached.					

Inputs/Outputs

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Inputs

PDR Review report

Requirements & architecture justification of solution elements (D&JF)

Solution design document (SSDD)

Solution elements' interface documents (ICD/IRS)

Solution elements' specifications (SSS/PIDS/SRS)

Outputs

Product Baseline (PBL)

Requirements & architecture justification of solution elements (D&JF)

Responsible

[Back to activity](#)

► Responsible ► Contributor

► Engineering Delivery Manager (EDM)	► Airworthiness Certification Engineer (ACM, DCS, RC...)
► Speciality Engineer (SPE)	► System Engineer (SE)
► System Engineering Manager (SysEM)	

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5.4 - CDR (Critical Design Review) (ref: 87202127-03)

Activity flow diagram



Presentation

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Objective

To confirm that the detailed design of the solution and its interfaces, including all its components, meets the solution requirements and confirms the completeness of the initial Product Baseline (PBL).

Description

For details see 87211053 Engineering Reviews instruction.

Documents

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TITLE / REFERENCE	TYPE	SCOPE	SPECIFIC	LANGUAGE	DATE
DDQS reviews checklists 87211132-DDQ-GRP-EN-018 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=120003684	TRS Template and Form		Hardware Software Solution	EN	27/06/2023
Engineering reviews instruction 87211053-DDQ-GRP-EN-006 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=119567360&vernum=9	Instruction			EN	11/07/2023
Export control Checklist 87215152-DDQ-GRP-EN-003 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=200146792	TRS Template and Form		Hardware Software Solution	EN	27/06/2023

Responsible

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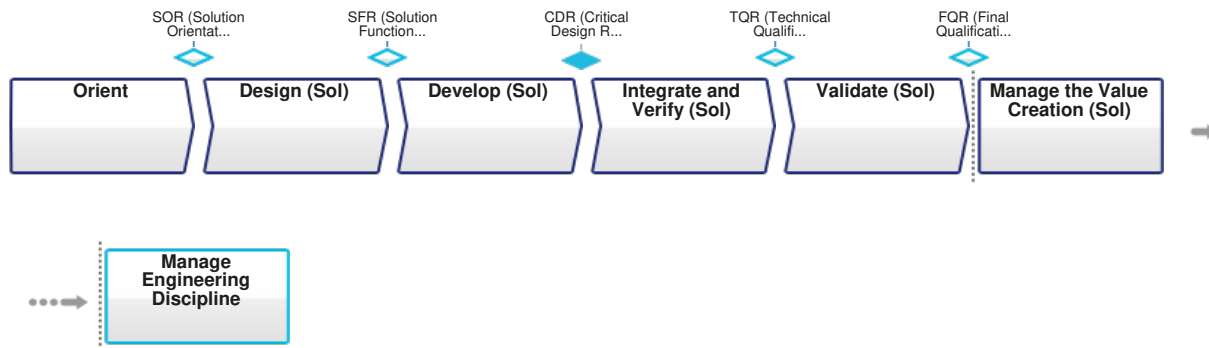
► Responsible ► Contributor

► Project Design Authority (PDA)	► Airworthiness Certification Engineer (ACM, DCS, RC...)
► Engineering Delivery Manager (EDM)	► Integration Verification Validation and Qualificat...
► MoB&F Manager	► Procurement Product/Bid/Project Manager (PPM)
► Production Process & Technology Manager (PPTM)	► Quality Assurance Manager (QAM)
► Service Engineering Manager (SvEM)	► System Engineering Manager (SysEM)

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6 - CDR (Critical Design Review) (ref: 87202127-03)

Activity flow diagram



Presentation

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Objective

To confirm that the detailed design of the solution and its interfaces, including all its components, meets the solution requirements and confirms the completeness of the initial Product Baseline (PBL).

Description

For details see 87211053 Engineering Reviews instruction.

Documents

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TITLE / REFERENCE	TYPE	SCOPE	SPECIFIC	LANGUAGE	DATE
DDQS reviews checklists 87211132-DDQ-GRP-EN-018 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=120003684	TRS	Template and Form	Hard ware Softw are Soluti on	EN	27/06/2023
Engineering reviews instruction 87211053-DDQ-GRP-EN-006 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=119567360&vernum=9		Instru ction		EN	11/07/2023
Export control Checklist 87215152-DDQ-GRP-EN-003 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=200146792	TRS	Template and Form	Hard ware Softw are Soluti on	EN	27/06/2023

Responsible

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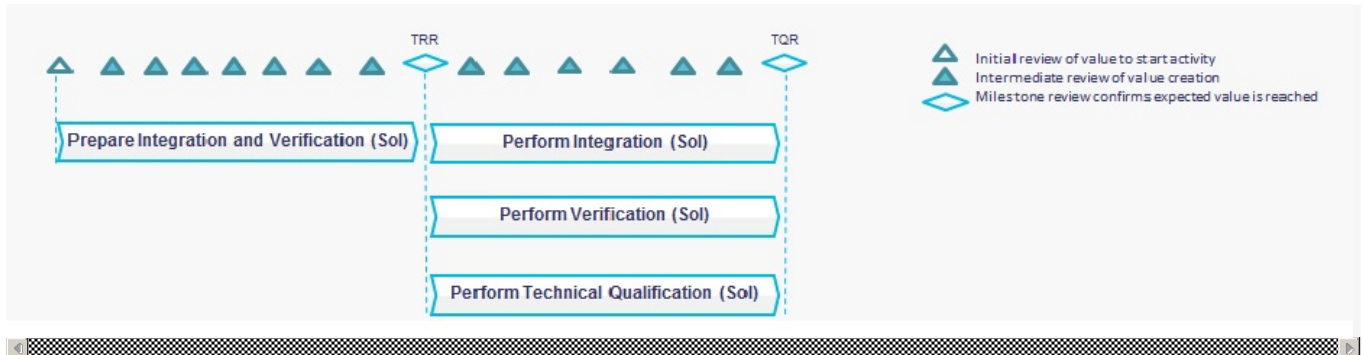
► Responsible ► Contributor

► Project Design Authority (PDA)	► Airworthiness Certification Engineer (ACM, DCS, RC...
► Engineering Delivery Manager (EDM)	► Integration Verification Validation and Qualificat...
► MoB&F Manager	► Procurement Product/Bid/Project Manager (PPM)
► Production Process & Technology Manager (PPTM)	► Quality Assurance Manager (QAM)
► Service Engineering Manager (SvEM)	► System Engineering Manager (SysEM)

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7 - Integrate and Verify (Sol) (ref: 87201336-00)

Activity flow diagram



Presentation

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Objective

To evaluate the design, behavior, interactions and performance of the solution or solution element and to confirm through provision of objective evidence that the requirements against which it has been designed are fulfilled.

Description

The above activities may be considered as sequenced or concurrent depending on the chosen development cycle.

Documents

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TITLE / REFERENCE	TYPE	SCOPE	SPECIFIC	LANGUAGE	DATE
Cybersecurity Engineering Guide 87210647-DDQ-GRP-EN_005 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=85758845	TRS	Guide line		EN	27/06/2023
Engineering Tests documents writing guide 87216829-DDQ-GRP-EN-003 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=226828035	TRS	Guide line	Hard ware Softw are Soluti on	EN	27/06/2023
Guide for vulnerability assessment during IVVQ activities 87215508-DDQ-GRP-EN-001 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=200149122	TRS	Guide line	Soluti on	EN	28/07/2020
IVVQP template 87210737-DDQ-GRP-EN-005 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=84659592	TRS	Templ ate and Form		EN	27/06/2023
Security engineering instruction 87210206-DDQ-GRP-EN-005 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=96019973&vernum=11		Instru ction		EN	27/06/2023
UX Design in Engineering 87216348-DDQ-GRP-EN-001 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=218307077&vernum=3		Instru ction	Soluti on	EN	28/09/2021

Roles

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No role for the selected object

Log of changes Group (inherited from: *Integrate and Verify (Sol)*)[Back to activity](#)

RELEASE	DESCRIPTION	DATE
I.2.2.0	CR059955: addition of the Airworthiness Certification Engineer (ACM, DCS, RCE) as contributor on Integrate & Verify the Solution activities	27/10/2020
F.2.0.5	CR046162 Replacement of the complete reference of the Engineering reviews instruction by the root of the document in the reviews description	27/09/2017
F.2.0.0	CR036118: Rework on structure and content to converge to analog activities on Sw/Hw as far as possible Addition of a "prepare I&V" as Sys-Em and ISO 15288 Inclusion of mandatory practices.	29/03/2017
E.2.0.2	CR 35056 : Correction of activity names (not unique)	30/03/2016
E.2.0.0	CR030615 :Update for installation in the following activities : TRR and Perform integration	27/01/2016
C.2.0.5	CR5149 :- precisions for incremental/evolutive development cycles- replacement of engineering disciplines role by System Engineer (SE)	24/09/2014
C.2.0.0	Clean up	29/01/2014
B.2.0.0	Cleanup, correction of coconsistency issues (CIICR-783)	26/09/2012
A.2.0.7	The D&JF French Instruction has been published (CIICR-607)	25/01/2012
A.2.0	Interface update (CIICR-346)	07/06/2011
A.1.0	CIICR-90, CIICR- 216, CIICR-219	15/04/2011
1.0	CHORUS 2.0 Go-live	10/03/2011

Note: Subject of the changes of the documents are detailed in the [Release History page](#).

RESPONSIBLE	
ACTORS	NAME AND ROLE
Content responsible	
Quality responsible	

7.1 - Prepare Integration and Verification (Sol) (ref: 87201336-06)

Activity flow diagram

Presentation

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Objective

To provide integration and verification test data, environments and resources in accordance with the IVVQ strategy.

Description

Characterise and plan integration and verification : integration logic and test campaigns (objective, schedule, closure criteria), test specifications, overall conformity matrix.

Develop tests to efficiently address the test campaigns objectives.

Receive the expected solution elements then verify their conformity to test coverage objective and availability of requested documents, data and tools.

Establish the availability and adequacy of the required resources (specialist, sub-contractor, support, etc.).

Establish and maintain representative environments.

Involve HW & SW disciplines to ensure completeness and efficiency

Documents

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TITLE / REFERENCE	TYPE	SCOPE	SPECIFIC	LANGUAGE	DATE
DIVVA Core Model Description for IVVQ Management 87211648-DDQ-GRP-EN-005 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=119867105	Guide line			EN	16/09/2022
Engineering Tests documents writing guide 87216829-DDQ-GRP-EN-003 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=226828035	Guide line		Hard ware Softw are Soluti on	EN	27/06/2023
IVVQ Handbook 87100124-DDQ-GRP-EN-009 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=19448365	Guide line			EN	27/06/2023
IVVQP template 87210737-DDQ-GRP-EN-005 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=84659592	Templ ate and Form			EN	27/06/2023
Solution Engineering with Agile 87211549-DDQ-GRP-EN-003 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=128280233	Guide line			EN	27/06/2023
Test Automation Capability Grid Template 87213511-DDQ-GRP-EN-001 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=166854381	Templ ate and Form			EN	11/12/2018

Inputs/Outputs

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Inputs

IVVQ Strategy

Outputs

Integration and verification environment

Integration and verification plan (IVVQP)

Integration procedures (STD)

Verification procedures (STD)

Responsible

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► Responsible

► Contributor

▸ Integration Verification Validation and Qualificat...	▸ Airworthiness Certification Engineer (ACM, DCS, RC...
▸ Configuration Manager	▸ Engineering Delivery Manager (EDM)
▸ Production Process & Technology Manager (PPTM)	▸ System Engineer (SE)
▸ System Engineering Manager (SysEM)	

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[Presentation](#) | [Context](#) | [Rules](#) | [Documents](#) | [Tools](#) | [Inputs / Outputs](#) | [Roles](#) | [Lifecycle](#) | [Tailoring](#) | [Performance criteria](#)

7.2 - TRR (Test Readiness Review) (ref: 87201336-01)

Activity flow diagram



Presentation

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Objective

To verify that all the necessary conditions for conducting the integration, verification and technical qualification activities at the appropriate level are met.

Description

For details see 87211053 DDQS reviews instruction.

Documents

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TITLE / REFERENCE	TYPE	SCOPE	SPECIFIC	LANGUAGE	DATE
DDQS reviews checklists 87211132-DDQ-GRP-EN-018 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=120003684	TRS Template and Form		Hardware Software Solution	EN	27/06/2023
Engineering reviews instruction 87211053-DDQ-GRP-EN-006 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=119567360&vernum=9	Instruction			EN	11/07/2023
Export control Checklist 87215152-DDQ-GRP-EN-003 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=200146792	TRS Template and Form		Hardware Software Solution	EN	27/06/2023

Responsible

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► Responsible ► Contributor

► Project Design Authority (PDA)	► Airworthiness Certification Engineer (ACM, DCS, RC...)
► Deployment Manager	► Engineering Delivery Manager (EDM)
► Integration Verification Validation and Qualificat...	► Production Process & Technology Manager (PPTM)
► Quality Assurance Manager (QAM)	► Service Engineering Manager (SvEM)
► System Engineering Manager (SysEM)	

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[Presentation](#) | [Context](#) | [Rules](#) | [Documents](#) | [Tools](#) | [Inputs / Outputs](#) | [Roles](#) | [Lifecycle](#) | [Tailoring](#) | [Performance criteria](#)

7.3 - Perform integration (Sol) (ref: 87201336-02)

Activity flow diagram

Presentation

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Objective

To assemble the solution or solution element in a logical and controlled manner, and to evaluate its design, behavior, interactions and performance, in accordance with the integration plan

Description

For each integration campaign:

- Record the configuration of each element received for integration (in factory or on site) from development production, or installation internal or external supplier; also record the configuration of the integration environment.
- Check elements against their configuration description, in particular the interfaces.
- Assemble the solution elements.
- Evaluate the solution behaviour, interactions and performance.
- Upon deviation related to interfaces descriptions, raise problem reports and when needed associated requests for deviation.
- Run the campaign rigorously, i.e. in accordance with its objectives and completion criteria
- Involve HW & SW disciplines in system Integration, Verification and Validation to ensure completeness and efficiency

Documents

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TITLE / REFERENCE	TYPE	SCOPE	SPECIFIC	LANGUAGE	DATE
TRIS DIVVA Core Model Description for IVVQ Management 87211648-DDQ-GRP-EN-005 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=119867105	Guide line			EN	16/09/2022
TRIS Engineering Tests documents writing guide 87216829-DDQ-GRP-EN-003 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=226828035	Guide line		Hard ware Softw are Soluti on	EN	27/06/2023
TRIS Guide for vulnerability assessment during IVVQ activities 87215508-DDQ-GRP-EN-001 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=200149122	Guide line		Soluti on	EN	28/07/2020
TRIS SUM template 87210731-DDQ-GRP-EN-003 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=85123257	Templ ate and Form		Softw are	EN	30/11/2021

Inputs/Outputs

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Inputs

Integration and verification environment
 Integration and verification plan (IVVQP)
 Integration procedures (STD)
 Solution elements
 TRR Review Report
 Verification procedures (STD)

Outputs

Integrated solution
 Integration Test Results (STR)
 PCRs

Responsible

[Back to activity](#)

► Responsible ► Contributor

► Integration Verification Validation and Qualificat...	► Airworthiness Certification Engineer (ACM, DCS, RC...
► Configuration Manager	► Engineering Delivery Manager (EDM)

▶ Production Process & Technology Manager (PPTM)	▶ Speciality Engineer (SPE)
▶ System Engineer (SE)	▶ System Engineering Manager (SysEM)

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7.4 - Perform verification (Sol) (ref: 87201336-03)

Activity flow diagram

Presentation

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Objective

To confirm through provision of objective evidence that the requirements against which the solution or solution element has been designed are fulfilled.

Description

For each verification campaign:

- Record the configuration of the solution or solution element and of its verification environment.
- Run the verification procedures and compile the outcomes (including the results).
- Record verification actual results and raise and investigate problem reports when deviating from the expected results.
- Update the overall conformity matrix with regard to verification results.
- Run the campaign rigorously, i.e. in accordance with its objectives and completion criteria
- Involve HW & SW disciplines in system Integration, Verification and Validation to ensure completeness and efficiency.

Documents

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TITLE / REFERENCE	TYPE	SCOPE	SPECIFIC	LANGUAGE	DATE
Engineering Tests documents writing guide 87216829-DDQ-GRP-EN-003 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=226828035	TRS	Guide line	Hard ware Softw are Soluti on	EN	27/06/2023
HUMAN_SYSTEMS_INTEGRATION_GUIDE 87203104-DDQ-GRP-EN-004 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=19447145	TRS	Guide line		EN	16/09/2022
SVDD template 87212328-DDQ-GRP-EN-003 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=136069565	TRS	Templ ate and Form	Hard ware Soluti on	EN	30/11/2021

Inputs/Outputs

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Inputs

Integrated solution
 Integration and verification environment
 Integration and verification plan (IVVQP)
 TRR Review Report
 Verification test procedures (STD)

Outputs

PCRs
 Verification Test Results (STR)
 Verified solution

Responsible

[Back to activity](#)

► Responsible ► Contributor

► Integration Verification Validation and Qualificat...	► Airworthiness Certification Engineer (ACM, DCS, RC...
► Configuration Manager	► Engineering Delivery Manager (EDM)
► Speciality Engineer (SPE)	► System Engineer (SE)
► System Engineering Manager (SysEM)	

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7.5 - Perform technical qualification (Sol) (ref: 87201336-04)

Activity flow diagram



Presentation

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Objective

To assess the readiness of solution elements before launching the validation and final qualification of the solution.

Description

Plan and participate in the various suppliers' Functional Configuration Audits (FCA) and Physical Configuration Audits (PCA) for components and/or solution elements.

Review integration and verification results and ensure that all requirements have been proven, and that any non-conformance has been recorded with associated requests for deviation when needed.

Analyse and approve the overall conformity matrix.

Ensure that the product data packages of solution elements are up to date for release to production.

Provide data and services for enabling systems required to produce, support or dispose of the solution.

Documents

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TITLE / REFERENCE	TYPE	SCOPE	SPECIFIC	LANGUAGE	DATE
EU declaration of conformity 87202844-DDQ-GRP-EN-005 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=119726744	TRS Template and Form			EN	30/11/2021
Instruction for producing Design & Justification File (D&JF) 87201508-DDQ-GRP-EN-009 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=96091671&vernum=12	Instruction		Hardware Software Solution	EN	28/11/2023 U!

Inputs/Outputs

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Inputs

Integration Test Results (STR)

PCRs

Verification Test Results (STR)

Outputs

Conformity matrix (D&JF)

Engineering data for in service and Support

FCA and PCA results of the solution elements

PCRs status

Requests for deviation

Technically qualified data package

Responsible

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► Responsible ► Contributor

► Integration Verification Validation and Qualificat...	► Airworthiness Certification Engineer (ACM, DCS, RC...
► Configuration Manager	► Engineering Delivery Manager (EDM)
► Production Process & Technology Manager (PPTM)	► Service Engineering Manager (SvEM)
► System Engineer (SE)	► System Engineering Manager (SysEM)

7.6 - TQR (Technical Qualification Review) (ref: 87202127-04)

Activity flow diagram



Presentation

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Objective

To confirm that integration and verification activities are complete with objective evidence that the solution meets the requirements (apart from operational environment).

To confirm that the product data packages of solution elements are accurate and detailed enough for release to production.

Description

For details see 87211053 DDQS reviews instruction.

Documents

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TITLE / REFERENCE	TYPE	SCOPE	SPECIFIC	LANGUAGE	DATE
DDQS reviews checklists 87211132-DDQ-GRP-EN-018 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=120003684	TRS	Template and Form	Hardware Software Solution	EN	27/06/2023
Engineering reviews instruction 87211053-DDQ-GRP-EN-006 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=119567360&vernum=9		Instruction		EN	11/07/2023
Export control Checklist 87215152-DDQ-GRP-EN-003 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=200146792	TRS	Template and Form	Hardware Software Solution	EN	27/06/2023
STrP template 87212329-DDQ-GRP-EN-002 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=136070399	TRS	Template and Form	Software	EN	30/11/2021

Responsible

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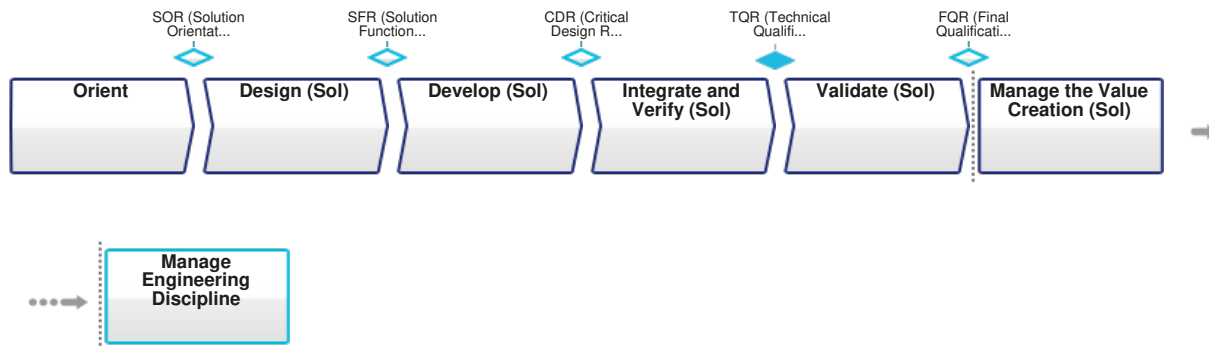
► Responsible ► Contributor

► Project Design Authority (PDA)	► Airworthiness Certification Engineer (ACM, DCS, RC...)
► Engineering Delivery Manager (EDM)	► Integration Verification Validation and Qualificat...
► Procurement Product/Bid/Project Manager (PPM)	► Production Process & Technology Manager (PPTM)
► Quality Assurance Manager (QAM)	► Service Engineering Manager (SvEM)
► System Engineering Manager (SysEM)	

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8 - TQR (Technical Qualification Review) (ref: 87202127-04)

Activity flow diagram



Presentation

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Objective

To confirm that integration and verification activities are complete with objective evidence that the solution meets the requirements (apart from operational environment).
 To confirm that the product data packages of solution elements are accurate and detailed enough for release to production.

Description

For details see 87211053 DDQS reviews instruction.

Documents

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TITLE / REFERENCE	TYPE	SCOPE	SPECIFIC	LANGUAGE	DATE
DDQS reviews checklists 87211132-DDQ-GRP-EN-018 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=120003684	TRS	Template and Form	Hardware Software Solution	EN	27/06/2023
Engineering reviews instruction 87211053-DDQ-GRP-EN-006 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=119567360&vernum=9		Instruction		EN	11/07/2023
Export control Checklist 87215152-DDQ-GRP-EN-003 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=200146792	TRS	Template and Form	Hardware Software Solution	EN	27/06/2023
STrP template 87212329-DDQ-GRP-EN-002 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=136070399	TRS	Template and Form	Software	EN	30/11/2021

Responsible

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▶ Responsible ▶ Contributor

▶ Project Design Authority (PDA)	▶ Airworthiness Certification Engineer (ACM, DCS, RC...
▶ Engineering Delivery Manager (EDM)	▶ Integration Verification Validation and Qualificat...
▶ Procurement Product/Bid/Project Manager (PPM)	▶ Production Process & Technology Manager (PPTM)
▶ Quality Assurance Manager (QAM)	▶ Service Engineering Manager (SvEM)
▶ System Engineering Manager (SysEM)	

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9 - Validate (Sol) (ref: 87201337-00)

Activity flow diagram



Presentation

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Objective

To demonstrate with objective evidence that the solution fulfills its intended use when placed in its intended environment as defined in the contract (or any other formal agreement), to support the certification (if required) and, from industrial point of view, to confirm that all IVV activities are completed and that the Product Data Package is ready for release to production and to support.

Description

The above steps or activities may be considered as sequenced, overlapped or concurrent depending on the chosen development cycle.

Documents

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TITLE / REFERENCE	TYPE	SCOPE	SPECIFIC	LANGUAGE	DATE
Cybersecurity Engineering Guide 87210647-DDQ-GRP-EN_005 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=85758845	Guide line			EN	27/06/2023
Security engineering instruction 87210206-DDQ-GRP-EN-005 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=96019973&vernum=11	Instru ction			EN	27/06/2023
UX Design in Engineering 87216348-DDQ-GRP-EN-001 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=218307077&vernum=3	Instru ction		Soluti on	EN	28/09/2021

Roles

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No role for the selected object

Log of changes Group (inherited from: *Validate (Sol)*)

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RELEASE	DESCRIPTION	DATE
I.2.2.0	CR059955: addition of the Airworthiness Certification Engineer (ACM, DCS, RCE) as contributor on Validate the Solution activities	27/10/2020
F.2.0.5	CR046162 Replacement of the complete reference of the Engineering reviews instruction by the root of the document in the reviews description	27/09/2017
F.2.0.0	CR036118: Addition of a "Prepare validation" for consistency with Sys-Em and ISO 15288 Inclusion of mandatory practices.	29/03/2017
E.2.0.0	CR030615: Update for installation in the following activities : TRR and Perform integration	27/01/2016

C.2.0.5	CR5149 :- precisions for incremental/evolutive development cycles- replacement of engineering disciplines role by System Engineer (SE)- rectification of validation objectives to be in line with SysEM	24/09/2014
C.2.0.0	Minor corrections (spelling, correction or addition of Inputs/Outputs, layout...)	29/01/2014
B.2.0.0	Cleanup, correction of consistency issues (CIICR-783)	26/09/2012
A.2.0.7	The D&JF French Instruction has been published (CIICR-607)	25/01/2012
A.2.0	Interface update (CIICR-346)	07/06/2011
A.1.0	CIICR-90, CIICR- 216, CIICR-219	15/04/2011
1.0	CHORUS 2.0 Go-live	10/03/2011

Note: Subject of the changes of the documents are detailed in the [Release History page](#).

RESPONSIBLE	
ACTORS	NAME AND ROLE
Content responsible	
Quality responsible	

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9.1 - Prepare Validation (ref: 87201337-05)

Activity flow diagram

Presentation

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Objective

To provide validation data, environments and resources in accordance with the IVVQ strategy.

Description

Characterize and plan validation : test campaigns (objective, schedule, closure criteria), test specifications, overall conformity matrix.

Define and verify validation procedures considering operational concepts (CONEMP, CONOPS, CONUSE) and possible reuse of validation assets.

Establish and verify the validation environment as the availability and adequacy of the required resources (customer, sub-contractor, etc.)

Obtain customer approval on work products according to the contract.

Involve HW & SW disciplines in system Integration, Verification and Validation to ensure completeness and efficiency

Implement an overall conformity matrix (VCRM)

Documents

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TITLE / REFERENCE	TYPE	SCOPE	SPECIFIC	LANGUAGE	DATE
TRIS DIVVA Core Model Description for IVVQ Management 87211648-DDQ-GRP-EN-005 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=119867105	Guide line			EN	16/09/2022
TRIS Engineering Tests documents writing guide 87216829-DDQ-GRP-EN-003 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=226828035	Guide line		Hard ware Softw are Soluti on	EN	27/06/2023
TRIS HUMAN_SYSTEMS_INTEGRATION_GUIDE 87203104-DDQ-GRP-EN-004 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=19447145	Guide line			EN	16/09/2022
TRIS IVVQP template 87210737-DDQ-GRP-EN-005 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=84659592	Templ ate and Form			EN	27/06/2023

Inputs/Outputs

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Inputs

IVVQ Strategy

Outputs

Validation environments

Validation plan (IVVQP)

Validation procedures (STD)

Responsible

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► Responsible ► Contributor

► Integration Verification Validation and Qualificat...	► Airworthiness Certification Engineer (ACM, DCS, RC...
► Configuration Manager	► Engineering Delivery Manager (EDM)
► Production Process & Technology Manager (PPTM)	► System Engineer (SE)
► System Engineering Manager (SysEM)	

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9.2 - TRR (Test Readiness Review) (ref: 87201336-01)

Activity flow diagram



Presentation

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Objective

To verify that all the necessary conditions for conducting the integration, verification and technical qualification activities at the appropriate level are met.

Description

For details see 87211053 DDQS reviews instruction.

Documents

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TITLE / REFERENCE	TYPE	SCOPE	SPECIFIC	LANGUAGE	DATE
DDQS reviews checklists 87211132-DDQ-GRP-EN-018 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=120003684	TRR Template and Form		Hardware Software Solution	EN	27/06/2023
Engineering reviews instruction 87211053-DDQ-GRP-EN-006 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=119567360&vernum=9	Instruction			EN	11/07/2023
Export control Checklist 87215152-DDQ-GRP-EN-003 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=200146792	TRR Template and Form		Hardware Software Solution	EN	27/06/2023

Responsible

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► Responsible ► Contributor

► Project Design Authority (PDA)	► Airworthiness Certification Engineer (ACM, DCS, RC...)
► Deployment Manager	► Engineering Delivery Manager (EDM)
► Integration Verification Validation and Qualificat...	► Production Process & Technology Manager (PPTM)
► Quality Assurance Manager (QAM)	► Service Engineering Manager (SvEM)
► System Engineering Manager (SysEM)	

9.3 - Perform solution validation (ref: 87201337-01)

Activity flow diagram

Presentation

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Objective

To acquire objective evidence that the solution fulfills its intended use when placed in its intended environment, as defined in the contract (or any other formal agreement). By executing the validation procedures in accordance with the validation plan.

Description

For each validation campaign:

- Record the configuration of the solution and of its validation environment.
- Run the validation procedures and compile the outcomes (including the results).
- Record validation actual results and raise and investigate problem reports when deviating from the expected results.
- Update the overall conformity matrix with regard to validation results.
- Run a test campaign rigorously, i.e. in accordance with its objectives and completion criteria
- Involve HW & SW disciplines in system Integration, Verification and Validation to ensure completeness and efficiency

Documents

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TITLE / REFERENCE	TYPE	SCOPE	SPECIFIC	LANGUAGE	DATE
TRIS DIVVA Core Model Description for IVVQ Management 87211648-DDQ-GRP-EN-005 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=119867105	Guide line			EN	16/09/2022
TRIS Engineering Tests documents writing guide 87216829-DDQ-GRP-EN-003 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=226828035	Guide line		Hard ware Softw are Soluti on	EN	27/06/2023
TRIS SVDD template 87212328-DDQ-GRP-EN-003 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=136069565	Templ ate and Form		Hard ware Soluti on	EN	30/11/2021

Inputs/Outputs

[Back to activity](#)

Inputs

TQR Report
 TRR Review Report
 Validation environments
 Validation plan (IVVQP)
 Validation procedures (STD)
 Verified solution

Outputs

PCRs
 Validated solution
 Validation Test Results (STR)

Responsible

[Back to activity](#)

► Responsible ► Contributor

► Integration Verification Validation and Qualificat...	► Airworthiness Certification Engineer (ACM, DCS, RC...
► Configuration Manager	► Deployment Manager
► Engineering Delivery Manager (EDM)	► Speciality Engineer (SPE)
► System Engineer (SE)	► System Engineering Manager (SysEM)

[Table of contents](#) > [Design, Develop and Qualify the Solution](#) > [Validate \(Sol\)](#) > **Perform solution qualification**

[Presentation](#) | [Context](#) | [Rules](#) | [Documents](#) | [Tools](#) | [Inputs / Outputs](#) | [Roles](#) | [Lifecycle](#) | [Tailoring](#) | [Performance criteria](#)

9.4 - Perform solution qualification (ref: 87201337-02)

Activity flow diagram



Presentation

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Objective

To demonstrate that all IVV activities are completed and that the Product Data Package is ready for release to production and to support.

Description

Produce the expected data (e.g. test data, manuals, etc.) for release to production and to support.

Collect solution elements/components configuration audits evidences (FCA,PCA).

Review validation results and ensure that all needs baseline have been proven, and that any non-conformance has been recorded with associated requests for deviation when needed.

Analyze and approve the overall conformity matrix.

Documents

[Back to activity](#)

TITLE / REFERENCE	TYPE	SCOPE	SPECIFIC	LANGUAGE	DATE
SVDD template 87212328-DDQ-GRP-EN-003 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=136069565	TRS	Template and Form	Hardware Solution	EN	30/11/2021

Inputs/Outputs

[Back to activity](#)

Inputs

Manufacturing and Control data

PCRs

Qualification plan (IVVQP)

TQR Report

TRR Review Report

Validated solution

Validation Test Results (STR)

Outputs

Conformity matrix (D&JF)

Engineering data for in service and Support

FCA and PCA results of the solution elements

PCRs status

Qualified Data Package (PBL)

Requests for deviation

Responsible

[Back to activity](#)

► Responsible ► Contributor

► Integration Verification Validation and Qualificat...	► Airworthiness Certification Engineer (ACM, DCS, RC...
► System Engineer (SE)	► System Engineering Manager (SysEM)

[Table of contents](#) > [Design, Develop and Qualify the Solution](#) > [Validate \(Sol\)](#) > **FQR (Final Qualification Review)**

[Presentation](#) | [Context](#) | [Rules](#) | [Documents](#) | [Tools](#) | [Inputs / Outputs](#) | [Roles](#) | [Lifecycle](#) | [Tailoring](#) | [Performance criteria](#)

9.5 - FQR (Final Qualification Review) (ref: 87202127-05)

Activity flow diagram



Presentation

[Back to activity](#)

Objective

To confirm that:

- All Design, Development and Qualification activities are complete.
- The PBL accurately documents the configuration of the solution.
- The solution (or solution element) fits its final use and is ready for production, deployment, transition to support, and disposal.

Description

For details see 87211053 Engineering Reviews instruction.

Documents

[Back to activity](#)

TITLE / REFERENCE	TYPE	SCOPE	SPECIFIC	LANGUAGE	DATE
DDQS reviews checklists 87211132-DDQ-GRP-EN-018 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=120003684	TRS	Template and Form	Hardware Software Solution	EN	27/06/2023
Engineering reviews instruction 87211053-DDQ-GRP-EN-006 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=119567360&vernum=9		Instruction		EN	11/07/2023
Export control Checklist 87215152-DDQ-GRP-EN-003 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=200146792	TRS	Template and Form	Hardware Software Solution	EN	27/06/2023
STrP template 87212329-DDQ-GRP-EN-002 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=136070399	TRS	Template and Form	Software	EN	30/11/2021

Responsible

[Back to activity](#)

► Responsible ► Contributor

► Project Design Authority (PDA)	► Airworthiness Certification Engineer (ACM, DCS, RC...)
► Engineering Delivery Manager (EDM)	► Integration Verification Validation and Qualificat...
► Production Process & Technology Manager (PPTM)	► Quality Assurance Manager (QAM)
► Service Engineering Manager (SvEM)	► System Engineering Manager (SysEM)

[Table of contents](#) > [Design, Develop and Qualify the Solution](#) > [Validate \(Sol\)](#) > **Support certification and acceptance**

[Presentation](#) | [Context](#) | [Rules](#) | [Documents](#) | [Tools](#) | [Inputs / Outputs](#) | [Roles](#) | [Lifecycle](#) | [Tailoring](#) | [Performance criteria](#)

9.6 - Support certification and acceptance (ref: 87201337-03)

Activity flow diagram



Presentation

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Objective

To contribute to the solution certification ('when required) and acceptance.

Description

Plan the engineering certification and acceptance supporting tasks.

Gather certification evidences (declaration of design and performance, conformity reports, etc.).

Participate in acceptance and certification tests specification, description, approbation, performance, and results analysis.

Participate in acceptance and certification reviews when expected.

Documents

[Back to activity](#)

TITLE / REFERENCE	TYPE	SCOPE	SPECIFIC	LANGUAGE	DATE
No document attached.					

Inputs/Outputs

[Back to activity](#)

Inputs

Acceptance plan and procedure

Certification plan

Outputs

Agreed deviations and waivers

Conformity matrix (D&JF)

Engineering data for Acceptance review

Engineering data for Certificates

Responsible

[Back to activity](#)

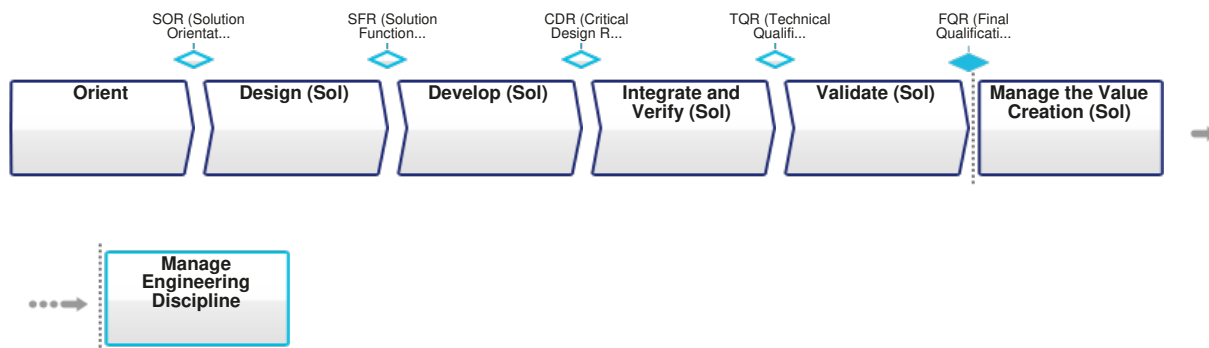
► Responsible ► Contributor

► Engineering Delivery Manager (EDM)	► Airworthiness Certification Engineer (ACM, DCS, RC...)
► Integration Verification Validation and Qualificat...	► Project Design Authority (PDA)
► System Engineer (SE)	► System Engineering Manager (SysEM)

[Table of contents](#) > [Design, Develop and Qualify the Solution](#) > **FQR (Final Qualification Review)**
[Presentation](#) | [Context](#) | [Rules](#) | [Documents](#) | [Tools](#) | [Inputs / Outputs](#) | [Roles](#) | [Lifecycle](#) | [Tailoring](#) | [Performance criteria](#)

10 - FQR (Final Qualification Review) (ref: 87202127-05)

Activity flow diagram



Presentation

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Objective

To confirm that:

- All Design, Development and Qualification activities are complete.
- The PBL accurately documents the configuration of the solution.
- The solution (or solution element) fits its final use and is ready for production, deployment, transition to support, and disposal.

Description

For details see 87211053 Engineering Reviews instruction.

Documents

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TITLE / REFERENCE	TYPE	SCOPE	SPECIFIC	LANGUAGE	DATE
DDQS reviews checklists 87211132-DDQ-GRP-EN-018 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=120003684	TRS	Template and Form	Hardware Software Solution	EN	27/06/2023
Engineering reviews instruction 87211053-DDQ-GRP-EN-006 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=119567360&vernum=9		Instruction		EN	11/07/2023
Export control Checklist 87215152-DDQ-GRP-EN-003 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=200146792	TRS	Template and Form	Hardware Software Solution	EN	27/06/2023
STrP template 87212329-DDQ-GRP-EN-002 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=136070399	TRS	Template and Form	Software	EN	30/11/2021

Responsible

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► Responsible ► Contributor

► Project Design Authority (PDA)	► Airworthiness Certification Engineer (ACM, DCS, RC...)
► Engineering Delivery Manager (EDM)	► Integration Verification Validation and Qualificat...
► Production Process & Technology Manager (PPTM)	► Quality Assurance Manager (QAM)
► Service Engineering Manager (SvEM)	► System Engineering Manager (SysEM)

[Table of contents](#) > [Design, Develop and Qualify the Solution](#) > **Manage the Value Creation (Sol)**
[Presentation](#) | [Context](#) | [Rules](#) | [Documents](#) | [Tools](#) | [Inputs / Outputs](#) | [Roles](#) | [Lifecycle](#) | [Tailoring](#) | [Performance criteria](#)

11 - Manage the Value Creation (Sol) (ref: 87211231-00)

Activity flow diagram



Presentation

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Objective

To make sure we deliver the expected value all across the solution lifecycle to the customer/market and to the company.

Description

These activities are performed concurrently with other engineering activities.

Documents

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TITLE / REFERENCE	TYPE	SCOPE	SPECIFIC	LANGUAGE	DATE
Cybersecurity Engineering Guide 87210647-DDQ-GRP-EN_005 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=85758845	TRS	Guide line		EN	27/06/2023
Engineering Management Plan (EMP) writing guide 87216448-DDQ-GRP-EN-006 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=215704008	TRS	Guide line	Hard ware Softw are Soluti on	EN	27/06/2023
Essential lessons learnt data requirements 87212432-QCI-GRP-EN-005 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=142323680&vernum=9		Instru ction		EN	29/03/2023
Safety Assurance Plan 87201699-DDQ-GRP-EN-006 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=276419367	TRS	Guide line	Hard ware Softw are Soluti on	EN	27/06/2023

Responsible

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► Responsible ► Contributor

► Engineering Work Package Manager (EWPM) ► MoB&F Manager

Log of changes Group (inherited from: *Manage the Value Creation (Sol)*)

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RELEASE	DESCRIPTION	DATE
L.2.2.0	CR080339: The Engineering Work Package Manager (EWPM) is added as contributor	28/11/2023

L.2.1.0	CR078999: Transfer of the "0130: Safety Assurance" from QCI to DDQ activities on "Plan the value creation" CR074804: The Safety Assurance Manager (SAM) is now part of DDQS process and contributes to the Plan the Value Creation activities	27/06/2023
F.2.0.0	CR036118: Simplification by reducing it to two activities with renaming and description review.	29/03/2017

Note: Subject of the changes of the documents are detailed in the [Release History page](#).

RESPONSIBLE	
ACTORS	NAME AND ROLE
Content responsible	
Quality responsible	

11.1 - Plan the Value Creation (Sol) (ref: 87211231-01)

Activity flow diagram



Presentation

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Objective

To organise and plan engineering activities with the objective of creating value for our customers/market and the company.

- Value for customers:
 - Added value for their mission execution, at an affordable price and on time
 - Best user experience during the whole product lifecycle
 - Deliver solutions that users can trust in all circumstances
- Value for the company:
 - Deliver the expected level of profitability
 - Create reusable assets in consistency with product policy
 - Continuously capitalize and share knowledge

Description

Identify & qualify the key value attributes (features, performances, quality attributes, ...) that drive the value for customers/the market and for the company.

In order to maximise the created value :

- Tailor the engineering process, practices and tools.
- Plan the value to be delivered during the development life cycle (roadmap, release plan with associated acceptance criteria).
- Organise and plan the engineering activities (WBS, OBS..).
- Assess engineering costs and risks & opportunities.
- Define qualitative and quantitative data to capitalize (i.e. metrics, lessons learned, knowledge).

Describe all the above items in the engineering plans, including engineering specialties (e.g. RAMT, safety , security, etc).

Formalise and launch the workpackages.

Support the establishment of purchasing contracts with sub-contractors.

Rules

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0130: Safety Assurance

When Safety requirements are identified, and depending on rules (delegation, threshold) which may be defined within the entity, a Safety Assurance Manager (or equivalent) shall be appointed for the bid, project or product, to define and implement a Safety Assurance Plan (or equivalent). Refer to the instruction "Safety Assurance" (87211146).

Documents

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TITLE / REFERENCE	TYPE	SCOPE	SPECIFIC	LANGUAGE	DATE	
Complexity Profiler Guide 87203444-DDQ-GRP-EN-007 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=88235717	TRS	Guide line		Solution	EN	27/06/2023
Complexity_Profiler template 87203605-DDQ-GRP-EN-004 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=88263544	TRS	Template and Form		Solution	EN	26/05/2020
Cybersecurity Engineering Guide 87210647-DDQ-GRP-EN_005 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=85758845	TRS	Guide line			EN	27/06/2023
Cybersecurity Guide for Equipment Architects 87214870-DDQ-GRP-EN-004 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=191297848	TRS	Guide line		Solution	EN	29/06/2022
DIVVA Core Model Description for IVVQ Management 87211648-DDQ-GRP-EN-005 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=119867105	TRS	Guide line			EN	16/09/2022
Development Life Cycle Guide 87201652-DDQ-GRP-EN-011 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=109677175	TRS	Guide line		Solution	EN	27/06/2023

TITLE / REFERENCE	TYPE	SCOPE	SPECIFIC	LANGU AGE	DATE
TRS Engineering Management Plan (EMP) writing guide 87216448-DDQ-GRP-EN-006 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=215704008	Guide line		Hard ware Softw are Soluti on	EN	27/06/2023
Engineering Metrics and KPIs for a Project 87204906-DDQ-GRP-EN-006 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=96088730&vernum=9	Instru ction		Hard ware Softw are Soluti on	EN	26/10/2021
Estimate Instruction 87201474-PRJ-GRP-EN-005 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=96089505&vernum=9	Instru ction			EN	31/01/2019
TRS Feature Model Template 87209707-DDQ-GRP-EN-006 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=65640486	Templ ate and Form			EN	27/06/2023
TRS Feature Modelling Guide 87209145-DDQ-GRP-EN-001 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=57144308	Guide line			EN	01/12/2014
TRS Guide for Co-Engineering practice implementation 87206365-DDQ-GRP-EN-004 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=41267858	Guide line			EN	27/06/2023
TRS HUMAN_SYSTEMS_INTEGRATION_GUIDE 87203104-DDQ-GRP-EN-004 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=19447145	Guide line			EN	16/09/2022
TRS IVVQ Handbook 87100124-DDQ-GRP-EN-009 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=19448365	Guide line			EN	27/06/2023
TRS IVVQP template 87210737-DDQ-GRP-EN-005 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=84659592	Templ ate and Form			EN	27/06/2023
Manage a Work Package 87201255-PRJ-GRP-EN-005 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=96089647&vernum=7	Instru ction			EN	09/10/2018
Management & Use of external software components (COTS & OSS) 87202149-DDQ-GRP-EN-009 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=96091491&vernum=12	Instru ction			EN	27/06/2023
TRS Metrics & KPI Guide 87204366-DDQ-GRP-EN-013 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=21518062	Guide line		Hard ware Softw are Soluti on	EN	27/06/2023
Product Line Engineering (PLE) 87210086-DDQ-GRP-EN-008 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=96019975&vernum=10	Instru ction			EN	28/11/2023 U!
TRS Project Engineering Dashboard 87203024-DDQ-GRP-EN-010 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=109677389	Templ ate and Form		Hard ware Softw are Soluti on	EN	29/06/2021
RAMT Instruction 87205688-SER-GRP-EN-004 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=96724631&vernum=5	Instru ction			EN	29/01/2019
Risk & Opportunity Instruction for Bids & Projects 87201475-PRJ-GRP-EN-012 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=96089519&vernum=20	Instru ction			EN	28/11/2023 U!
Safety Assurance 87211146-DDQ-GRP-EN-004 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=271355589&vernum=1	Guide line			EN	25/04/2023

TITLE / REFERENCE	TYPE	SCOPE	SPECIFIC	LANGUAGE	DATE
Safety Engineering Instruction 87204905-DDQ-GRP-EN-007 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=96088738&vernum=9	Instruction		Hardware Software Solution	EN	27/06/2023
Safety_Engineering_Guidance 87206928-DDQ-GRP-EN-001 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=30372497	TRSGuide line			EN	22/06/2013
Value Pulled Engineering (VPE) methodological guide 87211289-DDQ-GRP-EN-002 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=125433331	TRSGuide line		Hardware Software Solution	EN	29/01/2019
WBS instruction 87201470-PRJ-GRP-EN-007 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=96089687&vernum=10	Instruction			EN	28/07/2020

Inputs/Outputs

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Inputs

IVVQ Strategy
Integrated Solution Development Strategy
Market & Customer Requirement Description (for SOR)
Solution validation (DVa)
Value proposition, Business model and Pricing guideline

Outputs

Cost estimates (NRC, RC)
Engineering R&O register
Engineering plan (SEMP)
Expected value evaluation
IVVQ plan (IVVQP)
Initial value evaluation (D&JF, Agile business value,...)
Releases' content
Schedule
Work Package Description (WPD)

Responsible

[Back to activity](#)

► Responsible ► Contributor

► Engineering Delivery Manager (EDM)

► Safety Assurance Manager (SAM)

11.2 - Monitor the Value Creation (Sol) (ref: 87211231-02)

Activity flow diagram

Presentation

[Back to activity](#)

Objective

To make sure the engineering activities create the expected value and take appropriate resulting actions when deviates from the plans.

To make created value visible to all stakeholders.

Description

Measure and analyse actual performance including created value related to key value attributes.

Manage the technical debt.

Organise and perform engineering reviews in consistency with the defined development cycle.

Record decisions and related knowledge (alternatives, trade-off, actions ...).

Reassess and share initial assumptions (i.e. reviews of value assessment grid).

Support purchasing contracts execution.

Synthesize status and trends in a project engineering dashboard.

Capitalise data as expected.

Documents

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TITLE / REFERENCE	TYPE	SCOPE	SPECIFIC	LANGUAGE	DATE
Complexity Profiler Guide 87203444-DDQ-GRP-EN-007 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=88235717	TRS	Guide line		Solution	EN 27/06/2023
Cybersecurity Engineering Guide 87210647-DDQ-GRP-EN_005 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=85758845	TRS	Guide line			EN 27/06/2023
DIVVA Core Model Description for IVVQ Management 87211648-DDQ-GRP-EN-005 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=119867105	TRS	Guide line			EN 16/09/2022
Engineering Offshoring 87218608-DDQ-GRP-EN-002 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=277128744	TRS	Guide line		Solution	EN 13/10/2023 NI
Feature Model Template 87209707-DDQ-GRP-EN-006 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=65640486	TRS	Template and Form			EN 27/06/2023
Feature Modelling Guide 87209145-DDQ-GRP-EN-001 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=57144308	TRS	Guide line			EN 01/12/2014
IVVQ Handbook 87100124-DDQ-GRP-EN-009 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=19448365	TRS	Guide line			EN 27/06/2023
Metrics & KPI Guide 87204366-DDQ-GRP-EN-013 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=21518062	TRS	Guide line		Hardware Software Solution	EN 27/06/2023
Project Engineering Dashboard 87203024-DDQ-GRP-EN-010	TRS	Template and Form		Hardware Software Solution	EN 29/06/2021

TITLE / REFERENCE	TYPE	SCOPE	SPECIFIC	LANGUAGE	DATE
https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=109677389					
Value Pulled Engineering (VPE) methodological guide 87211289-DDQ-GRP-EN-002	Guide line		Hard ware Softw are Soluti on	EN	29/01/2019
https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=125433331					

Inputs/Outputs

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Inputs

Cost estimates (NRC, RC)
Engineering R&O register
Engineering plan (SEMP)
Expected value evaluation
Releases' content
Schedule
Work Package Description (WPD)

Outputs

Delivered value evaluation
Project engineering dashboard
Work package progress reports

Responsible

[Back to activity](#)

► Responsible ► Contributor

► Engineering Delivery Manager (EDM)

12 - Manage Engineering Discipline (ref: 87202127-06)

Activity flow diagram



Presentation

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Objective

Continuously enhance organisation's competitiveness which is built on attractiveness, quality, lead time and costs of products and solutions.

Description

Define the vision of the engineering organization.

Ensure engineering competitiveness by defining and driving the competitiveness improvement plan (Engineering Performance Plan) and the maturity of his organization, based on business challenges and weaknesses found from competitiveness KPI and root cause analysis of problems found on bids and projects.

Provide engineers with an efficient Engineering Environment made of :

- An efficient workspace that allows collaboration and focus.
- Tailored and competitive Process, Practices and Tools and efficient IT infrastructure that runs the tools (provided by IS/IT organization).
- An efficient support and coaching organization to support the deployment of process, practices and tools.

Ensure knowledge capitalization and sharing.

Ensure skills recruitment, evaluation and development, including engineers career development.

Ensure accountability on bids, products and projects deliverables, including the best allocation of resources to bids, products and projects.

Develop the maturity of the organisation and the customer awareness in cooperation with the Engineering Quality Leader (EQL).

Deploy an engineering "Make or Buy & Footprint" methodology in order to contribute to the achievement of competitiveness objectives.

Documents

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TITLE / REFERENCE	TYPE	SCOPE	SPECIFIC	LANGUAGE	DATE
Engineering governance-Redirection 87210749-DDQ-GRP-EN-003 https://ecm.corp.thales/livelink/livelink.exe?func=ll&objAction=download&objId=119577451&vernum=4	Instruction			EN	23/03/2021

Inputs/Outputs

[Back to activity](#)

Outputs

Make or Buy & Footprint (MoB&F) analysis and related transfer plan

Organisation engineering dashboard

Organisation engineering performance plan

Responsible

[Back to activity](#)

► Responsible ► Contributor

► Competence Center Engineering Manager (CC EM)	► Head of Discipline (HoD)
► Head of Engineering Delivery (HoED)	► Head of Engineering Operations (HoEO)
► MoB&F Manager	► System Engineering Manager (SysEM)
► Engineering Delivery Manager (EDM)	► Engineering Quality Leader (EQL)
► Hardware Engineering Manager (HwEM)	► Software Engineering Manager (SwEM)